

UNIQUENESS OF THE STATIONARY LOGARITHMIC DLR STATE

ABSTRACT. Fix an inverse temperature $\beta > 0$. For a stationary simple point process P on $\mathbb{R}$ with intensity one, write $v_P(t) = \text{Var}_P N_{(0,t]}$. We prove that the canonical logarithmic Dobrushin–Lanford–Ruelle equations have at most one such solution satisfying

$$v_P(t) \leq C_P t \quad (t > 0), \quad \lim_{t \rightarrow \infty} \frac{v_P(t)}{t} = 0, \quad \int_1^\infty \frac{v_P(t)}{t^2} dt < \infty.$$

The proof compares the conditional distributions of two candidate processes with the same finite-dimensional Gibbs measure: a logarithmic gas with lattice exterior particles and a quadratic potential of vanishing strength. A concentration inequality for averages of consecutive gaps controls the difference of expectations by relative entropy per particle. The integrated variance bound controls the exterior-force contribution in probability, and the sublinear variance bound makes the quadratic displacement contribution tend to zero. We then derive the variance assumptions from finite renormalized logarithmic electric energy, using a stationary joint distribution of the configuration and an electric field and a covariance spectral estimate. With the energy normalization specified below, this identifies every stationary finite-energy canonical DLR solution with Sine_β .

1. INTRODUCTION AND STATEMENTS

The process Sine_β describes the microscopic bulk limit of one-dimensional logarithmic gases. We use its unit-intensity normalization throughout. Dereudre, Hardy, Leblé and Maïda [DHLM19] show that it satisfies canonical Dobrushin–Lanford–Ruelle equations and that every stationary finite-energy solution of those equations is number-rigid. Their Section 1.3 asks whether the stationary solution is unique. The conditional universality theorem of Kuijlaars and Miña-Díaz [KMD19] supplies a deterministic route at $\beta = 2$. For general β , Bourgade, Erdős and Yau [BEY12] develop a comparison method for local logarithmic Gibbs measures, together with strong particle-location estimates for convex analytic potentials.

The organizing idea of the argument below is to compare both candidate infinite-volume laws with one common Gibbs measure with lattice exterior and then subtract the two comparison estimates. No limit of the reference measure is needed. A static convexity estimate converts relative entropy into control of averaged gap statistics. The integrated variance bound gives tightness of the conditional squared exterior-force difference. Sublinear variance controls the quadratic displacement introduced by the additional potential. The comparison theorem is stated directly in terms of these count-variance hypotheses and is independent of the electric-field argument used later to verify them.

Assiotis [Ass26, Theorem 1.1] proves stationary uniqueness for every $\beta > 0$ in the same finite-energy class, without an ergodicity assumption. His process energy is exactly $(2\pi)^{-1} \mathbb{E}_P W$; Lemma 7.1 checks the definitions, and Remark 8.1 compares the DLR formulations. His proof combines relative-entropy comparison with a circular β ensemble, cyclic-gap transport, the circular Palm limit, and Palm inversion. The argument below shares the entropy and Palm framework and provides an alternative comparison proof of that finite-energy uniqueness result. Its reference is a Gibbs measure with lattice exterior and a weak quadratic potential, which cancels between the two candidates

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without identifying its limit. Theorem 1.1 isolates the discrepancy assumptions sufficient for this comparison.

1.1. Configurations, notation, and the DLR specification. Let $\text{Conf}(\mathbb{R})$ denote the space of locally finite simple counting measures on $\mathbb{R}$, with the Borel sigma-algebra of the vague topology. For $\gamma \in \text{Conf}(\mathbb{R})$ and a Borel set A , write

$$N_A(\gamma) = \gamma(A), \quad \gamma_A = \gamma|_A, \quad (\theta_a \gamma)(A) = \gamma(A + a) \quad (a \in \mathbb{R}).$$

Thus θ_a moves a point at a to the origin. A probability measure P on $\text{Conf}(\mathbb{R})$ is stationary if $(\theta_a)_\# P = P$ for every a . Here $T_\# \nu(A) = \nu(T^{-1}(A))$ denotes the pushforward of a measure ν under a measurable map T . For a probability measure, it is the distribution of $T(X)$ when X has law ν . We also write $\text{Law}_\nu(X)$ for the distribution of a random variable X under ν . It has intensity one if

$$\mathbb{E}_P N_A = \text{Leb}(A) \quad \text{for every bounded Borel set } A \subset \mathbb{R}. \quad (1.1)$$

Here Leb is Lebesgue measure, and δ_x denotes the unit point mass at x . We use $\mathbb{N} = \{1, 2, \dots\}$. We omit the argument γ from counts when the configuration is clear. For a probability measure ν , we use $\nu(f) = \int f d\nu$ and $\text{Var}_\nu f = \nu(f^2) - \nu(f)^2$. For a stationary intensity-one process, set

$$v_P(t) = \text{Var}_P N_{[0,t]} = \mathbb{E}_P[(N_{[0,t]} - t)^2], \quad t > 0. \quad (1.2)$$

For vectors, $|\cdot|$ is the Euclidean norm; Id_K is the $K \times K$ identity matrix. A generic constant C may change from one occurrence to the next. Constants carrying a subscript record additional dependencies. In estimates for $K \rightarrow \infty$, constants are independent of K and of the confinement parameter T unless stated otherwise; dependence on fixed β, r, G , or on a fixed candidate law, is allowed. We write $X_K = o_P(a_K)$ for $X_K/a_K \rightarrow 0$ in P -probability, and $x_+ = \max\{x, 0\}$. The notation $a_K \asymp b_K$ means $c_1 b_K \leq a_K \leq c_2 b_K$ for positive constants c_1, c_2 independent of K .

We specify the conditional distribution meant by the canonical DLR equations. Let $I = (a, b)$ be a bounded interval, let $n \geq 1$, and define

$$\Omega_{I,n} = \{x \in \mathbb{R}^n : a < x_1 < \dots < x_n < b\}.$$

For an exterior configuration ξ supported in I^c , choose a reference vector $y \in \Omega_{I,n}$ and define the relative Hamiltonian

$$\begin{aligned} \mathcal{H}_I^\xi(x; y) = & - \sum_{1 \leq i < j \leq n} \log \frac{|x_j - x_i|}{|y_j - y_i|} \\ & - \lim_{R \rightarrow \infty} \sum_{\substack{p \in \xi \\ |p| \leq R}} \sum_{i=1}^n \log \frac{|x_i - p|}{|y_i - p|}. \end{aligned} \quad (1.3)$$

The cutoff in this definition is symmetric about the spatial origin. Whenever a principal-value sum $\text{PV} \sum_p$ occurs below, it means this limit with $|p| \leq R$ in the coordinates then in use; any explicitly excluded particles are omitted before taking the limit. The relative energies and the corresponding normalized kernel are required to exist for almost every exterior and particle number arising under the process. The kernel is

$$\begin{aligned} \kappa_{I,n}^\beta(\xi; dx) &= \frac{e^{-\beta \mathcal{H}_I^\xi(x; y)}}{Z_{I,n}^\beta(\xi; y)} \mathbf{1}_{\Omega_{I,n}}(x) dx, \\ Z_{I,n}^\beta(\xi; y) &= \int_{\Omega_{I,n}} e^{-\beta \mathcal{H}_I^\xi(x; y)} dx \in (0, \infty). \end{aligned} \quad (1.4)$$

Changing y adds a constant independent of x to the Hamiltonian and therefore does not change the kernel. For $n = 0$, the kernel is the point mass on the empty configuration.

Writing $\mathcal{G}_I = \sigma(\gamma_{I^c}, N_I)$, the canonical DLR property at inverse temperature β is the identity

$$\mathbb{E}_P[f(\gamma) \mid \mathcal{G}_I] = \int f\left(\gamma_{I^c} + \sum_{i=1}^{N_I} \delta_{x_i}\right) \kappa_{I, N_I}^\beta(\gamma_{I^c}; dx) \quad P\text{-almost surely} \quad (1.5)$$

for every bounded measurable $f : \text{Conf}(\mathbb{R}) \rightarrow \mathbb{R}$ and every bounded interval I . This is the ordered-coordinate form of the relative-energy specification in [DHL19, Section 2.1].

Theorem 1.1 (Uniqueness under discrepancy control). *Fix $\beta > 0$. There is at most one stationary simple point process of intensity one satisfying the canonical logarithmic DLR equations and the following three conditions:*

$$v_P(t) \leq C_P t \quad (t > 0), \quad (1.6)$$

$$v_P(t) = o(t) \quad (t \rightarrow \infty), \quad (1.7)$$

$$\int_1^\infty v_P(t) t^{-2} dt < \infty. \quad (1.8)$$

The constant C_P need not be uniform over candidate processes.

For clarity, all three conditions are retained in the statement. Section 7 proves them together for each ergodic finite-energy DLR solution. The comparison proof itself requires no ergodicity.

Theorem 1.2 (Finite-energy uniqueness). *Fix $\beta > 0$, and let W be the renormalized logarithmic energy defined in (7.4), with the inner truncation and unit background specified in Section 7. If P is stationary, satisfies the canonical logarithmic DLR property (1.5), and $\mathbb{E}_P W < \infty$, then*

$$P = \text{Sine}_\beta.$$

Theorem 1.2 follows from Theorem 1.1, Proposition 7.6, the ergodic DLR decomposition in [DHL19, Proposition 3.13], and the finite-energy existence input of [LS17, Corollary 1.2]. The energy condition uses a background of intensity one. Mean intensity one by itself is insufficient for uniqueness: the equal mixture of the vacuum and the image of Sine_β under $x \mapsto x/2$ is a stationary canonical DLR solution of intensity one and infinite energy [Ass26, Proposition 9.4]. Indeed, if P_{mix} denotes that mixture and Λ its component intensity, then

$$P(\Lambda = 0) = P(\Lambda = 2) = \frac{1}{2}, \quad \text{Var}_{P_{\text{mix}}} N_{(0,t]} \geq \text{Var}(\Lambda t) = t^2.$$

It therefore also fails (1.6).

1.2. Organization and main estimates. Section 2 defines the finite-dimensional reference measure, the gap average, relative entropy, and relative Fisher information before stating the comparison inequalities (2.7) and (2.8). The reference has lattice exterior particles and a quadratic potential of strength T^{-1} . Section 3 collects the counting and Palm facts needed for random particle-ended intervals. Sections 4 and 5 control, respectively, the confinement cost and the exterior-force contribution. These estimates are assembled in Section 6. The electric-field argument in Section 7 verifies the hypotheses of Theorem 1.1 for finite-energy states.

The three variance hypotheses have distinct roles in the comparison:

Assumption	Main role
Linear bound (1.6)	Counting regularity and integrable Palm count envelopes (Section 3).
Sublinear growth (1.7)	Endpoint-length control and negligible quadratic displacement (Section 4).
Integrated variance (1.8)	Half-line discrepancy and exterior-force envelopes (Sections 3 and 5).

Proposition 7.6 collects the three conclusions for ergodic finite-energy components. Appendix A supplies the convex-domain and weak Sobolev justification used in the static comparison estimate.

2. A STATIC COMPARISON INEQUALITY FOR GAP STATISTICS

Fix $K \geq 1$, write $L = K + 1$, and let

$$\Omega_K = \{u \in \mathbb{R}^K : 0 < u_1 < \cdots < u_K < L\}.$$

Set $u_0 = 0$, $u_{K+1} = L$, and $g_i(u) = u_{i+1} - u_i$ for $0 \leq i \leq K$. The reference exterior is

$$\xi_K^{\text{lat}} = \{j \in \mathbb{Z} : j \leq 0 \text{ or } j \geq L\}.$$

With any fixed $r_* \in (0, L)$, define

$$H_K^{\text{lat}}(u) = - \sum_{1 \leq i < j \leq K} \log(u_j - u_i) - \sum_{i=1}^K \text{PV} \sum_{p \in \xi_K^{\text{lat}}} \log \left| \frac{u_i - p}{r_* - p} \right|. \quad (2.1)$$

Changing r_* changes only an additive constant. Euler's product for the gamma function gives the following exact finite expression:

$$H_K^{\text{lat}}(u) = - \sum_{1 \leq i < j \leq K} \log(u_j - u_i) + \sum_{i=1}^K \log \frac{\Gamma(u_i) \Gamma(L - u_i)}{\Gamma(r_*) \Gamma(L - r_*)}. \quad (2.2)$$

Here $\Gamma(z) = \int_0^\infty t^{z-1} e^{-t} dt$ for $z > 0$. The paired exterior series converges, and its derivatives of order at least two converge locally uniformly. The endpoint singularities are $-\log u_i - \log(L - u_i)$. For $T > 0$ put

$$\omega_{K,T}(du) = Z_{K,T}^{-1} e^{-\Phi_{K,T}(u)} du, \quad \Phi_{K,T}(u) = \beta H_K^{\text{lat}}(u) + \frac{1}{2T} \sum_{i=1}^K (u_i - i)^2. \quad (2.3)$$

Here and below du denotes K -dimensional Lebesgue measure, and

$$Z_{K,T} = \int_{\Omega_K} e^{-\Phi_{K,T}(u)} du \in (0, \infty). \quad (2.4)$$

Fix an integer $r \geq 1$ and a function $G \in C_c^2((0, \infty)^r)$. For $K > r$, define

$$F_K(u) = \frac{1}{K-r} \sum_{i=1}^{K-r} G(g_i(u), \dots, g_{i+r-1}(u)). \quad (2.5)$$

This uses only gaps between interior particles. Adding or deleting at most m edge windows, with m fixed, changes the normalized average by at most $2m \|G\|_\infty / (K - r - m)$ for $K > r + m$; this bound tends to zero as $K \rightarrow \infty$.

For probability measures $\mu \ll \omega$ on Ω_K , write $h = d\mu/d\omega$. The relative entropy and relative Fisher information are, respectively,

$$S(\mu \mid \omega) = \int h \log h d\omega, \quad I(\mu \mid \omega) = 4 \int \left| \nabla \sqrt{h} \right|^2 d\omega, \quad (2.6)$$

with $0 \log 0 = 0$. If $\mu \not\ll \omega$, both quantities are defined to be $+\infty$. Here $I = +\infty$ unless $\sqrt{h}$ belongs to the weighted weak Sobolev space specified in Appendix A. For positive locally differentiable densities this is $\int |\nabla \log h|^2 d\mu$.

Proposition 2.1 (Entropy and Fisher-information bounds for gap averages). *Fix $\beta > 0$, an integer $r \geq 1$, and $G \in C_c^2((0, \infty)^r)$. There is a constant $C_G < \infty$ such that, for every integer $K \geq 2r + 1$, every $T > 0$, and every probability measure μ on Ω_K ,*

$$|\mu(F_K) - \omega_{K,T}(F_K)|^2 \leq \frac{C_G}{K} S(\mu \mid \omega_{K,T}), \quad (2.7)$$

$$|\mu(F_K) - \omega_{K,T}(F_K)|^2 \leq \frac{C_G T}{K} I(\mu \mid \omega_{K,T}). \quad (2.8)$$

More precisely, choose $B < \infty$ so that $\text{supp } G \subset (0, B)^r$, and put

$$M_0 = \|G\|_\infty, \quad M_1 = \max_a \|\partial_a G\|_\infty, \quad M_2 = \max_{a,b} \|\partial_{ab} G\|_\infty.$$

The constant may be chosen to depend only on $\beta, r, B, M_0, M_1, M_2$.

Proof. Write $\Phi = \Phi_{K,T}$ and $n = K - r \geq K/2$. For $v \in \mathbb{R}^K$, extend v by $v_0 = v_{K+1} = 0$ and put $d_i = v_{i+1} - v_i$. Nearest-neighbor interactions and the two endpoint charges give

$$v^\top \Phi''(u) v \geq \beta \sum_{i=0}^K \frac{d_i^2}{g_i(u)^2} + \frac{1}{T} |v|^2. \quad (2.9)$$

Every remaining logarithmic interaction has nonnegative Hessian. We use the stronger gap-dependent part of this lower bound to obtain the factor K^{-1} ; the uniform part alone would not give it.

First gap derivatives. Regard F_K as a function of the gaps and set $q_j = \partial F_K / \partial g_j$. Each gap belongs to at most r windows, so

$$q_j = \frac{1}{n} \sum_{\substack{1 \leq i \leq n \\ i \leq j \leq i+r-1}} \partial_{j-i+1} G(g_i, \dots, g_{i+r-1}), \quad |q_j| \leq \frac{r M_1}{n} \leq \frac{2r M_1}{K}.$$

The endpoint coefficients q_0, q_K vanish. Since the support of each derivative is contained in $\text{supp } G$, a nonzero q_j requires $g_j \leq B$. Consequently

$$\sum_{j=0}^K q_j^2 g_j^2 \leq \frac{4r^2 M_1^2 B^2}{K}.$$

Weighted Cauchy–Schwarz and (2.9) now give

$$|\nabla F_K \cdot v|^2 = \left| \sum_j q_j d_j \right|^2 \leq \left(\beta^{-1} \sum_j q_j^2 g_j^2 \right) \left(\beta \sum_j \frac{d_j^2}{g_j^2} \right) \leq \frac{A_1}{K} v^\top \Phi'' v, \quad A_1 = \frac{4r^2 B^2 M_1^2}{\beta}.$$

Taking the supremum over v in this quadratic-form duality yields

$$\nabla F_K^\top (\Phi'')^{-1} \nabla F_K \leq A_1 / K. \quad (2.10)$$

Second gap derivatives and window overlap. Let $w_i = (g_i, \dots, g_{i+r-1})$ and let J_i be the indicator that w_i belongs to the support of $D^2 G$. Then

$$\begin{aligned} |v^\top F_K'' v| &\leq \frac{M_2}{n} \sum_{i=1}^n J_i \left(\sum_{a=0}^{r-1} |d_{i+a}| \right)^2 \\ &\leq \frac{r M_2}{n} \sum_{i=1}^n J_i \sum_{a=0}^{r-1} d_{i+a}^2 \\ &\leq \frac{r M_2 B^2}{n} \sum_{i=1}^n \sum_{a=0}^{r-1} \frac{d_{i+a}^2}{g_{i+a}^2} \leq \frac{r^2 M_2 B^2}{n} \sum_{j=0}^K \frac{d_j^2}{g_j^2}. \end{aligned}$$

In the third line, every gap in an active window is at most B ; in the last line, every gap is counted at most r times. Thus

$$|v^\top F_K'' v| \leq \frac{A_2}{K} v^\top (\beta H_K^{\text{lat}}) v, \quad A_2 = \frac{2r^2 B^2 M_2}{\beta}. \quad (2.11)$$

Uniformity over the tilted laws. For $s \in \mathbb{R}$, define the exponentially tilted probability measure

$$\omega_s(du) = \frac{e^{sF_K(u)}}{\omega_{K,T}(e^{sF_K})} \omega_{K,T}(du),$$

Set $c_G = [2(1 + A_2)]^{-1}$. The same pointwise estimate (2.11), with no dependence on s, K, T , gives for every $|s| \leq c_G K$

$$(\Phi - sF_K)'' \geq \frac{1}{2}\beta H_K^{\text{lat}} + T^{-1}\text{Id}_K \geq \frac{1}{2}\Phi''.$$

The Brascamp–Lieb inequality in Lemma A.1, followed by (2.10), therefore gives

$$\text{Var}_{\omega_s} F_K \leq \mathbb{E}_{\omega_s} [\nabla F_K^\top ((\Phi - sF_K)'')^{-1} \nabla F_K] \leq \frac{2A_1}{K} \quad (|s| \leq c_G K).$$

This estimate first holds on each gap-truncated simplex and passes to ω_s with the same constants, as proved in the appendix. If $\psi(s) = \log \omega_{K,T}(e^{sF_K})$, then $\psi''(s) = \text{Var}_{\omega_s} F_K$. Integrating twice from zero yields

$$\log \mathbb{E}_{\omega_{K,T}} \exp\{s(F_K - \omega_{K,T}(F_K))\} \leq A_1 s^2 / K, \quad |s| \leq c_G K. \quad (2.12)$$

Entropy and Fisher information. The entropy inequality, used with either sign, gives

$$|\mu(F_K) - \omega_{K,T}(F_K)| \leq \frac{S(\mu \mid \omega_{K,T})}{s} + \frac{Ds}{K}, \quad 0 < s \leq c_G K, \quad D = 1 + A_1.$$

Abbreviate $S = S(\mu \mid \omega_{K,T})$ for the rest of this paragraph. When $0 < S \leq Dc_G^2 K$, choose $s = \sqrt{SK/D}$ to obtain the square bound $4DS/K$; the case $S = 0$ follows by $s \downarrow 0$. When $S > Dc_G^2 K$, use $|\mu(F_K) - \omega_{K,T}(F_K)| \leq 2M_0$ instead. Thus (2.7) holds, for example, with any

$$C_G \geq \max\{4D, 4M_0^2/(Dc_G^2)\}.$$

When $S = +\infty$, (2.7) is immediate. Finally, the uniform convexity in (2.9) and the logarithmic Sobolev part of Lemma A.1 imply

$$S(\mu \mid \omega_{K,T}) \leq \frac{T}{2} I(\mu \mid \omega_{K,T}), \quad (2.13)$$

which proves (2.8) after enlarging the constant if necessary. The static convexity inequalities are also used in [BEY12, Sections 3 and 5]. The appendix makes their use on the singular simplex, including the weak Fisher-information domain, explicit for every $\beta > 0$. $\square$

3. DISCREPANCY AND PALM PRELIMINARIES

Throughout Sections 3–6, P satisfies Theorem 1.1. Its Palm distribution P^0 is the probability measure on configurations with a point at zero characterized by Campbell's identity

$$\mathbb{E}_P \sum_{p \in \gamma} f(p, \theta_p \gamma) = \int_{\mathbb{R}} \mathbb{E}_{P^0} f(x, \gamma) dx \quad (3.1)$$

for every nonnegative measurable f on $\mathbb{R} \times \text{Conf}(\mathbb{R})$. For a rooted configuration with infinitely many points in each direction, enumerate its points and gaps as

$$\cdots < p_{-1} < p_0 = 0 < p_1 < \cdots, \quad g_j = p_{j+1} - p_j \quad (j \in \mathbb{Z}). \quad (3.2)$$

The successor shift is $T_*\gamma = \theta_{p_1}\gamma$; thus $g_j(T_*\gamma) = g_{j+1}(\gamma)$. Palm measure is invariant under this map: $(T_*)_{\#}P^0 = P^0$. The almost-sure existence of the two-sided enumeration follows from the next lemma.

Lemma 3.1 (Almost-sure exterior regularity). *There is a translation-invariant measurable set $\mathcal{R}$ with $P(\mathcal{R}) = P^0(\mathcal{R}) = 1$ such that, for every $\gamma \in \mathcal{R}$ and every $a \in \mathbb{R}$,*

$$\lim_{t \rightarrow \infty} \frac{N_{(a, a+t]}(\gamma)}{t} = \lim_{t \rightarrow \infty} \frac{N_{[a-t, a)}(\gamma)}{t} = 1, \quad (3.3)$$

and

$$\int_1^\infty \frac{(N_{(a, a+t]} - t)^2 + (N_{[a-t, a)} - t)^2}{t^2} dt < \infty. \quad (3.4)$$

Moreover,

$$\lim_{R \rightarrow \infty} \sum_{\substack{p \in \gamma \\ 0 < |p-a| \leq R}} \frac{1}{p-a} \quad \text{exists as a finite real number for every } a \in \mathbb{R}. \quad (3.5)$$

A point at a , when present, is omitted from this sum.

Proof. By (1.6), Chebyshev's inequality along $t = n^2$ and Borel–Cantelli give $N_{(0,t]}/t \rightarrow 1$; monotonic interpolation removes the subsequence. The backward statement is identical. Tonelli and (1.8) give (3.4) at $a = 0$. Define the oriented cumulative discrepancy by

$$D_0(x) = \begin{cases} N_{(0,x]} - x, & x \geq 0, \\ -N_{(x,0]} - x, & x < 0. \end{cases} \quad (3.6)$$

Then $N_{(a, a+t]} - t = D_0(a+t) - D_0(a)$ for every $t > 0$. The backward discrepancy with the convention $[a-t, a)$ differs from $D_0(a) - D_0(a-t)$ by at most two endpoint indicators. A change of variable and comparability of t and $a+t$ for large t prove the integrability at every finite a , simultaneously on the same event.

For the principal value, put $D_{\text{sym}}(t) = N_{(0,t]} - N_{[-t,0]}$. Then

$$\sum_{1 < |p| \leq R, p \in \gamma} \frac{1}{p} = \frac{D_{\text{sym}}(R)}{R} - D_{\text{sym}}(1) + \int_1^R \frac{D_{\text{sym}}(t)}{t^2} dt.$$

The first term vanishes by density one. The integral converges absolutely by Cauchy–Schwarz and (3.4). Add the finite sum over $0 < |p| \leq 1$. The same argument at any a gives (3.5). All the resulting properties hold simultaneously at every a , so their common event is translation invariant. Equation (3.1) transfers this event to Palm measure. $\square$

Lemma 3.2 (Integrable bounds for Palm counting functions). *Under P^0 define*

$$C^+ = \sup_{t \geq 0} \frac{N_{(0,t]}}{1+t}, \quad C^- = \sup_{t \geq 0} \frac{N_{[-t,0]}}{1+t}.$$

Then

$$\mathbb{E}_{P^0} C^+ + \mathbb{E}_{P^0} C^- < \infty. \quad (3.7)$$

Proof. For a spatially rooted sample put $C_0 = \sup_{t \geq 0} N_{(0,t]}/(1+t)$. A dyadic bound gives

$$C_0 \leq N_{(0,1]} + 2 + \sum_{j \geq 0} 2^{-j} \left| N_{(0, 2^{j+1}]} - 2^{j+1} \right|.$$

Minkowski's inequality and (1.6) show $\mathbb{E}_P C_0^2 < \infty$. For every $p \in \gamma \cap (0, 1]$, $C^+(\theta_p \gamma) \leq 2C_0(\gamma)$. Campbell's formula at intensity one therefore gives

$$\mathbb{E}_{P^0} C^+ = \mathbb{E}_P \sum_{p \in \gamma \cap (0, 1]} C^+(\theta_p \gamma) \leq 2\mathbb{E}_P [N_{(0,1]} C_0] < \infty.$$

Reflect the argument for C^- . $\square$

3.1. The first positive point and the preceding gap. Use the Palm points and gaps in (3.2). Campbell's identity gives the following size-bias formula. If a is the first positive point of a spatial sample, the law of $\theta_a \gamma$ is

$$\tilde{P}^0(d\gamma) = g_{-1}(\gamma)P^0(d\gamma). \quad (3.8)$$

Indeed, each gap ending at a point p contributes the interval of possible spatial origins of length $g_{-1}(\theta_p \gamma)$. In particular $\mathbb{E}_{P^0} g_{-1} = 1$.

Lemma 3.3 (Uniform domination of endpoint environments). *Let a_j be the $(j+1)$ -st positive point of a spatial sample, and let $\mathfrak{g}_*$ be the length of the whole gap containing its spatial origin. For every $M > 0$, every integer $j \geq 0$, and every nonnegative measurable function Φ on rooted configurations,*

$$\mathbb{E}_P[\mathbf{1}_{\{\mathfrak{g}_* \leq M\}} \Phi(\theta_{a_j} \gamma)] \leq M \mathbb{E}_{P^0} \Phi. \quad (3.9)$$

Thus the restricted endpoint laws are dominated by MP^0 , uniformly in the number j of successor shifts.

Proof. The event $\{\mathfrak{g}_* \leq M\}$ becomes $\{g_{-1} \leq M\}$ when the sample is rooted at its first positive point. Since $\theta_{a_j} \gamma = T_*^j(\theta_{a_0} \gamma)$, (3.8) gives

$$\mathbb{E}_P[\mathbf{1}_{\{\mathfrak{g}_* \leq M\}} \Phi(\theta_{a_j} \gamma)] = \mathbb{E}_{P^0}[g_{-1} \mathbf{1}_{\{g_{-1} \leq M\}} \Phi(T_*^j \gamma)] \leq M \mathbb{E}_{P^0}[\Phi \circ T_*^j] = M \mathbb{E}_{P^0} \Phi.$$

The last equality uses $(T_*)_{\#} P^0 = P^0$. No invariance of the truncating event, and no independence of that event from the shifted environment, is required. $\square$

The next lemma addresses the bias in (3.8) without assuming ergodicity. The first positive point has the preceding-gap weighted law, so a pointwise ergodic limit of a block average need not itself be constant. What removes the weight is that the conditional mean of the preceding gap on every point-shift invariant component is one.

Lemma 3.4 (Spatial block averages identify Palm gap laws). *Let $r \geq 1$ be an integer and let $G : (0, \infty)^r \rightarrow \mathbb{R}$ be bounded and measurable. Let a_j be the $(j+1)$ -st positive point of a configuration sampled under P , for $j \geq 0$. For $K > r$, set*

$$A_{K,r}(\gamma) = \frac{1}{K-r} \sum_{i=1}^{K-r} G(a_{i+1} - a_i, \dots, a_{i+r} - a_{i+r-1}). \quad (3.10)$$

Then

$$\lim_{K \rightarrow \infty} \mathbb{E}_P A_{K,r} = \mathbb{E}_{P^0} G(g_0, \dots, g_{r-1}). \quad (3.11)$$

Proof. Put $\Psi(\gamma) = G(g_0, \dots, g_{r-1})$ and let $\mathcal{I}$ be the invariant sigma-algebra of T_* . Define the rooted average

$$A_K^0 = \frac{1}{K-r} \sum_{i=1}^{K-r} \Psi \circ T_*^i, \quad A_{K,r}(\gamma) = A_K^0(\theta_{a_0} \gamma).$$

Birkhoff's theorem gives $A_K^0 \rightarrow \mathbb{E}_{P^0}[\Psi \mid \mathcal{I}]$ P^0 -almost surely. Shifting the starting index by a fixed number changes a bounded average by $O(K^{-1})$. Density one and $\mathbb{E}_{P^0} g_{-1} < \infty$ give

$$\mathbb{E}_{P^0}[g_{-1} \mid \mathcal{I}] = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{j=0}^{n-1} g_{-1} \circ T_*^j = \lim_{n \rightarrow \infty} \frac{p_{n-1} - p_{-1}}{n} = 1.$$

Here the middle equality telescopes the gaps and the last uses the almost-sure particle density from Lemma 3.1. By (3.8), the expectation of the spatially rooted block average is $\mathbb{E}_{P^0}[g_{-1} A_K^0]$. Since $|A_K^0| \leq \|G\|_{\infty}$, dominated convergence with the integrable weight g_{-1} , followed by conditioning on $\mathcal{I}$, yields

$$\lim_{K \rightarrow \infty} \mathbb{E}_{P^0}[g_{-1} A_K^0] = \mathbb{E}_{P^0}[g_{-1} \mathbb{E}_{P^0}[\Psi \mid \mathcal{I}]] = \mathbb{E}_{P^0}[\mathbb{E}_{P^0}[g_{-1} \mid \mathcal{I}] \mathbb{E}_{P^0}[\Psi \mid \mathcal{I}]] = \mathbb{E}_{P^0} \Psi.$$

$\square$

4. DLR INTERVALS WITH A FIXED PARTICLE NUMBER

For a spatial sample let a be the first positive point and let b be the $(K+2)$ -nd positive point. The open interval (a, b) contains exactly K points, written $x_1 < \dots < x_K$. Put

$$L = K + 1, \quad \ell = b - a, \quad c = L/\ell, \quad u_i = c(x_i - a).$$

Let $\zeta = \gamma - \sum_{i=1}^K \delta_{x_i}$ denote the retained exterior. Both endpoints, all negative points, and all points beyond b are retained. The conditioning sigma-field is

$$\mathcal{G}_K = \sigma(\zeta) = \sigma\{\zeta(A) : A \subset \mathbb{R} \text{ bounded Borel}\}.$$

From ζ , the point a is recovered as its first positive point and b as its second positive point. On $\{b < R\}$ the original number of points in $(0, R)$ is $\zeta((0, R)) + K$. Thus the endpoints, the event $\{b < R\}$, and this original particle number on that event are all $\mathcal{G}_K$ -measurable.

Lemma 4.1 (DLR conditioning between particles of fixed ranks). *For every bounded measurable $f : \mathbb{R}^K \rightarrow \mathbb{R}$,*

$$\mathbb{E}_P[f(x_1, \dots, x_K) \mid \mathcal{G}_K] = \int_{\Omega_{(a,b),K}} f(x) \kappa_{(a,b),K}^\beta(\zeta; dx) \quad P\text{-almost surely.} \quad (4.1)$$

Let $S_{a,c}(x) = (c(x_1 - a), \dots, c(x_K - a))$ and let $\hat{\zeta} = (p \mapsto c(p - a))_\# \zeta$ be the scaled exterior. The conditional distribution of the rescaled coordinates is

$$\mu_\zeta = (S_{a,c})_\# \kappa_{(a,b),K}^\beta(\zeta; \cdot) = \kappa_{(0,L),K}^\beta(\hat{\zeta}; \cdot), \quad \mathbb{E}_P[f(u) \mid \mathcal{G}_K] = \mu_\zeta(f). \quad (4.2)$$

Its domain is Ω_K , and $\hat{\zeta}$ contains particles at 0 and L .

Proof. Fix a positive integer R and work on $A_R = \{b < R\}$. Canonical DLR in the deterministic interval $(0, R)$ first conditions on $\gamma|_{(0,R)^c}$ and on $n = N_{(0,R)}$. For $n \geq K+2$, write its ordered coordinates as $y_1 < \dots < y_n$. Further condition on

$$y_1, \quad y_{K+2}, y_{K+3}, \dots, y_n.$$

The free coordinates are exactly $y_2, \dots, y_{K+1}$, with domain $y_1 < y_2 < \dots < y_{K+1} < y_{K+2}$. In the deterministic-interval Gibbs density, all terms involving only retained particles are constant under this conditioning. The remaining terms are precisely the pair interactions among these K free particles and their interactions with every retained particle, including $y_1 = a$ and $y_{K+2} = b$. Disintegration of the ordered Lebesgue density therefore gives the normalized Gibbs kernel κ_ζ with exterior ζ .

More explicitly, on A_R the retained data in this two-stage conditioning generate the same sigma-field as ζ : the number n is $\zeta((0, R)) + K$, the retained coordinates are the points of ζ in $(0, R)$, and the deterministic exterior is also read from ζ . For bounded measurable h and f , disintegration on each particle-number sector consequently gives

$$\mathbb{E}_P[\mathbf{1}_{A_R} h(\zeta) f(x_1, \dots, x_K)] = \mathbb{E}_P[\mathbf{1}_{A_R} h(\zeta) \kappa_\zeta(f)].$$

Every permitted interior resampling leaves a the first positive point and b the $(K+2)$ -nd positive point. The rank-selection indicator is therefore identically one on this conditional domain; it contributes no further factor to the density. Finally $A_R \uparrow 1$ as integer $R \rightarrow \infty$, so bounded convergence proves the displayed conditional-expectation identity without its indicator. This identifies the law given $\mathcal{G}_K$.

The affine assertion follows from Lemma 4.2 below. The Jacobian is constant and is absorbed by normalization. $\square$

Lemma 4.2 (Affine covariance of relative exterior energies). *Let $I = (a, b)$ be a bounded interval with K interior particles and a locally finite exterior ζ of density one in both directions, for which the symmetric relative exterior energies exist. For $c > 0$, the map $S_{a,c} : x \mapsto c(x - a)$ satisfies*

$$(S_{a,c})_{\#} \kappa_{I,K}^{\beta}(\zeta; \cdot) = \kappa_{(0,c(b-a)),K}^{\beta}((p \mapsto c(p - a))_{\#} \zeta; \cdot). \quad (4.3)$$

The same symmetric-cutoff convention is used on both sides. More precisely, for the exterior summand $\Phi_p(x, y)$ defined in the proof,

$$\lim_{S \rightarrow \infty} \sup_{x, y \in I^K} \left| \sum_{\substack{p \in \zeta \\ |p-a| \leq S}} \Phi_p(x, y) - \sum_{\substack{p \in \zeta \\ |p| \leq S}} \Phi_p(x, y) \right| = 0. \quad (4.4)$$

Proof. For two interior configurations $x, y \in I^K$, the relative contribution of an exterior point p is, up to the fixed Hamiltonian sign,

$$\Phi_p(x, y) = \sum_{i=1}^K \log \frac{|x_i - p|}{|y_i - p|}.$$

For $|p| \geq 2(1 + |a| + |b|)$ the mean value theorem gives

$$\sup_{x, y \in I^K} |\Phi_p(x, y)| \leq \frac{2K(b-a)}{|p|}.$$

Compare cutoffs $|p| \leq S$ and $|p - a| \leq S$. Their symmetric difference is contained in

$$\mathcal{S}_S = \{p : S - |a| \leq |p| \leq S + |a|\}.$$

Density one in each direction implies $\zeta(\mathcal{S}_S) = o(S)$: the count on each fixed-width shell is bounded by the difference of two cumulative counts of the form $t + o(t)$. Hence

$$\sup_{x, y \in I^K} \left| \sum_{\substack{p \in \zeta \\ |p-a| \leq S}} \Phi_p(x, y) - \sum_{\substack{p \in \zeta \\ |p| \leq S}} \Phi_p(x, y) \right| \leq \frac{CK(b-a)}{S} \zeta(\mathcal{S}_S) = o(1).$$

The uniformity here is for fixed K, a, b , exactly as needed for each conditional law; no uniform rate in the random boundary is asserted. Writing $q = c(p - a)$, $u_i = c(x_i - a)$ and $v_i = c(y_i - a)$ gives

$$\log \frac{|u_i - q|}{|v_i - q|} = \log \frac{|x_i - p|}{|y_i - p|}.$$

A symmetric cutoff $|q| \leq R$ corresponds to $|p - a| \leq R/c$; the preceding shell bound identifies its limit with the original one. Relative pair energies are unchanged as well, since their $\log c$ terms cancel between x and y . Thus the transformed relative Hamiltonian agrees with that of the scaled exterior. The constant Jacobian c^{-K} cancels in the normalized density. In particular the endpoint particles are sent exactly to 0 and L in the rescaling used above. $\square$

Write $\mathfrak{g}_* = a - p_-$, where p_- is the last point before the spatial origin. This is the whole gap containing that origin. It is measurable from ζ , and $a \leq \mathfrak{g}_*$.

Lemma 4.3 (Concentration of the endpoint distance). *The endpoint length satisfies*

$$\forall \varepsilon > 0, \quad \lim_{K \rightarrow \infty} P(|\ell - L| > \varepsilon \sqrt{K}) = 0. \quad (4.5)$$

Equivalently, $\ell - L = o_P(\sqrt{K})$. For any two candidate processes P, Q , there is a deterministic sequence $r_K \downarrow 0$ such that, for each fixed M and

$$E_{K,M} = \{\mathfrak{g}_* \leq M, |\ell - L| \leq r_K \sqrt{K}\}, \quad (4.6)$$

$$\limsup_{K \rightarrow \infty} P(E_{K,M}^c) \leq P(\mathfrak{g}_* > M),$$

and the same assertion holds for Q . The event $E_{K,M}$ is retained-exterior measurable.

Proof. Fix $\varepsilon > 0$ and put $t_{\pm} = K + 2 \pm \varepsilon \sqrt{K}$. For all sufficiently large K , Chebyshev's inequality gives

$$P(b > t_+) \leq \frac{v_P(t_+)}{\varepsilon^2 K} \rightarrow 0, \quad P(b < t_-) \leq \frac{v_P(t_-)}{\varepsilon^2 K} \rightarrow 0,$$

because $t_{\pm}/K \rightarrow 1$ and (1.7) holds. Now

$$\frac{\ell - L}{\sqrt{K}} = \frac{b - (K + 2)}{\sqrt{K}} + \frac{1 - a}{\sqrt{K}} \rightarrow 0 \quad \text{in } P\text{-probability.}$$

The variable a is finite almost surely and does not depend on K . A diagonal choice gives a deterministic $r_K \downarrow 0$ such that

$$P(|\ell - L| > r_K \sqrt{K}) + Q(|\ell - L| > r_K \sqrt{K}) \rightarrow 0.$$

The assertions about $E_{K,M}$ follow. □

Define the conditional quadratic cost

$$J_K(\zeta) = \mathbb{E}_{\mu_\zeta} \sum_{i=1}^K (u_i - i)^2. \quad (4.7)$$

Lemma 4.4 (Subquadratic expected displacement). *For every fixed M ,*

$$K^{-2} \mathbb{E}_P[\mathbf{1}_{E_{K,M}} J_K] \rightarrow 0. \quad (4.8)$$

Proof. For deterministic ordered $z_1, \dots, z_K \in (0, \ell)$, let $A(t)$ be their counting function and $\rho = K/\ell$. Direct expansion yields the exact identity

$$\int_0^\ell (A(t) - \rho t)^2 dt = \rho \sum_{i=1}^K \left(z_i - \frac{i - \frac{1}{2}}{\rho} \right)^2 + \frac{\ell}{12}. \quad (4.9)$$

For example, $\int_0^\ell A(t)^2 dt = \sum_i (2i - 1)(\ell - z_i)$ and $\int_0^\ell t A(t) dt = \frac{1}{2} \sum_i (\ell^2 - z_i^2)$, which verify (4.9).

Take $z_i = x_i - a$ and use D_0 from (3.6). Except at endpoints of Lebesgue measure zero,

$$A(t) - \rho t = D_0(a + t) + (a - 1) + (1 - \rho)t.$$

On $E_{K,M}$, $a \leq M$, $\ell \asymp K$, and $b \leq 3K$ for all sufficiently large K (with M fixed). To compare the two deterministic particle locations, set $m_i = (i - \frac{1}{2})\ell/K$. Since $L = K + 1$,

$$\left| m_i - \frac{i\ell}{L} \right| = \frac{\ell}{K} \left| \frac{i}{L} - \frac{1}{2} \right| \leq \frac{\ell}{2K} \quad (1 \leq i \leq K).$$

Consequently, (4.9) yields

$$\sum_{i=1}^K (u_i - i)^2 \leq \frac{2c^2}{\rho} \int_0^\ell (A(t) - \rho t)^2 dt + \frac{L^2}{2K}.$$

Also,

$$\int_0^\ell (A(t) - \rho t)^2 dt \leq 3 \int_0^{3K} D_0(s)^2 ds + 3\ell(a-1)^2 + \ell(\ell-K)^2.$$

Here $c^2/\rho = L^2/(K\ell)$ is bounded on $E_{K,M}$, and $(\ell-K)^2 \leq 2(\ell-L)^2 + 2$. These inequalities give, on this event,

$$\sum_i (u_i - i)^2 \leq C \int_0^{3K} D_0(s)^2 ds + C_M K + CK(\ell-L)^2. \quad (4.10)$$

By (1.7), $\int_0^{3K} v_P(s) ds = o(K^2)$, and on $E_{K,M}$ the last term in (4.10) is at most $Cr_K^2 K^2$. Therefore

$$\frac{1}{K^2} \mathbb{E}_P \left[\mathbf{1}_{E_{K,M}} \sum_{i=1}^K (u_i - i)^2 \right] \leq \frac{C}{K^2} \int_0^{3K} v_P(s) ds + \frac{C_M}{K} + Cr_K^2 \longrightarrow 0.$$

Because the good event is retained-exterior measurable, conditional expectation turns the left side into the expression in (4.8). $\square$

5. TIGHTNESS OF THE CONDITIONAL EXTERIOR-FORCE CONTRIBUTION

The force estimates will only use tightness. We will not require an integrable product of an exterior energy and an interior count envelope.

Lemma 5.1 (A square-integrable bound for the half-line force). *Let q be a particle and let the exterior lie on one side of q . After deleting q , define its outward counting function by one of*

$$N_{\text{ext}}(t) = \gamma([q - t, q)) \quad \text{or} \quad N_{\text{ext}}(t) = \gamma((q, q + t]), \quad t \geq 0.$$

Assume the configuration belongs to the regularity set of Lemma 3.1. Define

$$D(t) = N_{\text{ext}}(t) - (t-1)_+, \quad H(x) = \int_0^\infty \frac{|D(t)|}{(x+t)^2} dt, \quad x \geq 0. \quad (5.1)$$

The function $(t-1)_+$ is the cumulative mass of the continuum ray $[1, \infty)$ with unit density. The associated bound satisfies

$$\mathcal{B} = H(0)^2 + \int_0^\infty H(x)^2 dx < \infty. \quad (5.2)$$

For $c \in [1/2, 2]$, define the rescaled quantities by

$$\begin{aligned} D^{(c)}(t) &= N_{\text{ext}}(t/c) - c^{-1}(t-1)_+, \\ H^{(c)}(x) &= \int_0^\infty \frac{|D^{(c)}(t)|}{(x+t)^2} dt, \\ \mathcal{B}^{(c)} &= H^{(c)}(0)^2 + \int_0^\infty H^{(c)}(x)^2 dx. \end{aligned} \quad (5.3)$$

There is an absolute constant $C < \infty$ such that

$$\sup_{1/2 \leq c \leq 2} \mathcal{B}^{(c)} \leq C(\mathcal{B} + 1). \quad (5.4)$$

Proof. Simplicity and local finiteness give $d \in (0, 1]$ such that no residual exterior point has outward distance less than d . Thus $D(t) = 0$ on $[0, d)$. Lemma 3.1 controls the tail, and Cauchy–Schwarz gives

$$\int_0^\infty \frac{D(t)^2}{t^2} dt < \infty, \quad H(0)^2 \leq \frac{1}{d} \int_0^\infty \frac{D(t)^2}{t^2} dt < \infty.$$

For completeness, the Schur-test kernel $K(x, t) = t/(x+t)^2$ satisfies

$$\int_0^\infty K(x, t) t^{-1/2} dt = \frac{\pi}{2} x^{-1/2}, \quad \int_0^\infty K(x, t) x^{-1/2} dx = \frac{\pi}{2} t^{-1/2}.$$

Since $H(x) = \int_0^\infty K(x, t) |D(t)|/t dt$, the Schur test gives

$$\int_0^\infty H(x)^2 dx \leq \frac{\pi^2}{4} \int_0^\infty D(t)^2 t^{-2} dt. \quad (5.5)$$

For the scaled configuration the centered discrepancy is

$$D(t/c) + (t/c - 1)_+ - c^{-1}(t - 1)_+.$$

Write the last two terms as $R_c(t) = (t/c - 1)_+ - c^{-1}(t - 1)_+$. For $c \in [1/2, 2]$,

$$|R_c(t)| \leq C \mathbf{1}_{\{t \geq 1/2\}}, \quad H^{(c)}(x) \leq c^{-1}H(x/c) + \frac{C}{x + 1/2}.$$

Squaring, integrating in x , and evaluating at $x = 0$ proves (5.4), with the same constant for every $c \in [1/2, 2]$. $\square$

Let $B_\zeta(u)$ be the difference of the actual and lattice exterior forces in the rescaled block. More precisely, with endpoint terms cancelled,

$$B_\zeta(u) = \text{PV} \sum_{p \in \widehat{\zeta} \setminus \{0, L\}} \frac{1}{u - p} - \text{PV} \sum_{p \in \xi_K^{\text{lat}} \setminus \{0, L\}} \frac{1}{u - p}, \quad 0 \leq u \leq L.$$

For almost every retained exterior under a candidate law, each paired sum exists and their difference is bounded on $[0, L]$. Define

$$\mathcal{D}_K(\zeta) = \mathbb{E}_{\mu_\zeta} \sum_{i=1}^K B_\zeta(u_i)^2. \quad (5.6)$$

Proposition 5.2 (Tightness of the conditional squared force difference). *For fixed M , the nonnegative random variables $\mathbf{1}_{E_{K,M}} \mathcal{D}_K$ are tight as $K \rightarrow \infty$. Explicitly,*

$$\lim_{A \rightarrow \infty} \limsup_{K \rightarrow \infty} P(E_{K,M} \cap \{\mathcal{D}_K > A\}) = 0. \quad (5.7)$$

The statement holds separately for each candidate process, with no assertion of a uniform first moment for $\mathcal{D}_K$.

Proof. On $E_{K,M}$, $c = L/\ell \in [1/2, 2]$ for all sufficiently large K . The scaled exterior $\widehat{\zeta}$ has density $\lambda = c^{-1} = \ell/L$. Define its left and right outward counts, centered discrepancies, and nonnegative bounds by

$$\begin{aligned} N_L(t) &= \widehat{\zeta}([-t, 0)), & N_R(t) &= \widehat{\zeta}((L, L+t]), \\ D_\star(t) &= N_\star(t) - \lambda(t-1)_+, & H_\star(x) &= \int_0^\infty \frac{|D_\star(t)|}{(x+t)^2} dt, \\ \mathcal{B}_\star &= H_\star(0)^2 + \int_0^\infty H_\star(x)^2 dx, & \star &\in \{L, R\}. \end{aligned} \quad (5.8)$$

Both endpoint particles have already been removed from these counts. For the residual unit lattice, set

$$D_{\text{lat}}(t) = \lfloor t \rfloor - (t-1)_+, \quad 0 \leq D_{\text{lat}}(t) \leq \mathbf{1}_{\{t \geq 1\}}.$$

Stieltjes integration by parts, first with finite cutoffs, gives the exact signed decomposition

$$\begin{aligned} B_\zeta(u) &= \int_0^\infty \frac{D_L(t) - D_{\text{lat}}(t)}{(u+t)^2} dt - \int_0^\infty \frac{D_R(t) - D_{\text{lat}}(t)}{(L-u+t)^2} dt \\ &\quad + (\lambda - 1) \log \frac{L-u+1}{u+1}. \end{aligned} \quad (5.9)$$

Indeed, the boundary terms at infinity vanish because $D_\star(t) = o(t)$, and the continuum contribution is

$$\int_1^\infty \left(\frac{1}{u+t} - \frac{1}{L-u+t} \right) dt = \log \frac{L-u+1}{u+1}.$$

The change from a common outward cutoff to the symmetric spatial cutoff has vanishing error, by the same fixed-width shell estimate as in Lemma 4.2. Finally,

$$\int_0^\infty \frac{D_{\text{lat}}(t)}{(x+t)^2} dt \leq \frac{1}{x+1}, \quad x \geq 0.$$

Consequently,

$$\begin{aligned} |B_\zeta(u)| &\leq H_L(u) + H_R(L-u) + \frac{C}{u+1} + \frac{C}{L-u+1} \\ &\quad + |\lambda - 1| \left| \log \frac{u+1}{L-u+1} \right|. \end{aligned} \quad (5.10)$$

The endpoint charges at $0, L$ cancel exactly before this estimate. The quantities $\mathcal{B}_L$ and $\mathcal{B}_R$ in (5.8) depend only on the retained exterior.

For an interior configuration $u \in \Omega_K$, define

$$C_{\text{int}}^L(u) = \sup_{t \geq 0} \frac{\#\{i : u_i \leq t\}}{1+t}, \quad C_{\text{int}}^R(u) = \sup_{t \geq 0} \frac{\#\{i : L - u_i \leq t\}}{1+t}. \quad (5.11)$$

For any nonnegative nonincreasing function $h : [0, L] \rightarrow [0, \infty)$, Stieltjes integration by parts gives

$$\sum_i h(u_i)^2 \leq C_{\text{int}}^L \left(h(0)^2 + \int_0^L h(s)^2 ds \right). \quad (5.12)$$

For instance, set $N_{\text{int}}(t) = \#\{i : u_i \leq t\}$ and use $N_{\text{int}}(t) \leq C_{\text{int}}^L(1+t)$ in $\int h^2 dN_{\text{int}}$; the terminal boundary term gives exactly the expression on the right. The right-endpoint version is

$$\sum_{i=1}^K h(L - u_i)^2 \leq C_{\text{int}}^R \left(h(0)^2 + \int_0^L h(s)^2 ds \right).$$

For $h_L(s) = \log((L+1)/(s+1))$, one has

$$\left| \log \frac{u+1}{L-u+1} \right| \leq h_L(u) + h_L(L-u), \quad h_L(0)^2 + \int_0^L h_L(s)^2 ds \leq CL.$$

Squaring (5.10), using (5.12), and taking conditional expectations gives

$$\mathcal{D}_K \leq C\{X_L(\mathcal{B}_L + 1) + X_R(\mathcal{B}_R + 1)\} + C(\lambda - 1)^2 L(X_L + X_R), \quad (5.13)$$

where the exterior-measurable conditional expectations are

$$X_\star(\zeta) = \mu_\zeta(C_{\text{int}}^\star), \quad \star \in \{L, R\}. \quad (5.14)$$

We verify (5.7) under the distribution of the retained exterior induced by P , in four steps. All bounds below are for fixed M and sufficiently large K .

Exterior envelopes. Let $\mathcal{B}^{\text{out},-}$ and $\mathcal{B}^{\text{out},+}$ be the unscaled leftward and rightward quantities of Lemma 5.1, viewed as functions of a Palm-rooted configuration. They are finite P^0 -almost surely, but no expectation bound for them is needed. The actual left endpoint is $a = a_0$, and the right endpoint is $b = a_{K+1}$. The scale estimate in that lemma implies on $E_{K,M}$

$$\mathcal{B}_L \leq C\{\mathcal{B}^{\text{out},-}(\theta_a \gamma) + 1\}, \quad \mathcal{B}_R \leq C\{\mathcal{B}^{\text{out},+}(\theta_b \gamma) + 1\}.$$

Both right sides depend only on the appropriate outward half-line. Using Lemma 3.3, uniformly in K ,

$$\begin{aligned} P(E_{K,M} \cap \{\mathcal{B}_L > A\}) &\leq MP^0(\mathcal{B}^{\text{out},-} > A/C - 1), \\ P(E_{K,M} \cap \{\mathcal{B}_R > A\}) &\leq MP^0(\mathcal{B}^{\text{out},+} > A/C - 1). \end{aligned}$$

The right sides tend to zero as $A \rightarrow \infty$. The domination constant at the right endpoint is unchanged although its point-shift index $K + 1$ grows with K .

Conditional count envelopes. For an actual sample on the good event, $c \in [1/2, 2]$ gives

$$C_{\text{int}}^L \leq 2C^+(\theta_a \gamma), \quad C_{\text{int}}^R \leq 2C^-(\theta_b \gamma).$$

For example $\#\{i : u_i \leq t\} \leq N_{(a,a+t/c]}(\gamma) \leq C^+(\theta_a \gamma)(1 + t/c)$, and $(1 + t/c)/(1 + t) \leq 2$. By Lemma 4.1, $X_\star = \mathbb{E}_P[C_{\text{int}}^\star | \mathcal{G}_K]$. Since $E_{K,M}$ belongs to $\mathcal{G}_K$, Lemmas 3.3 and 3.2 imply

$$\begin{aligned} \mathbb{E}_P[\mathbf{1}_{E_{K,M}} X_L] &= \mathbb{E}_P[\mathbf{1}_{E_{K,M}} C_{\text{int}}^L] \leq 2M\mathbb{E}_{P^0} C^+, \\ \mathbb{E}_P[\mathbf{1}_{E_{K,M}} X_R] &= \mathbb{E}_P[\mathbf{1}_{E_{K,M}} C_{\text{int}}^R] \leq 2M\mathbb{E}_{P^0} C^-. \end{aligned} \tag{5.15}$$

The bound is uniform in K . Markov's inequality therefore makes $\mathbf{1}_{E_{K,M}} X_\star$ uniformly tight.

Products. For $\star \in \{L, R\}$ and any $A > 0$, the union bound gives

$$\begin{aligned} P(E_{K,M} \cap \{X_\star(\mathcal{B}_\star + 1) > A\}) &\leq P(E_{K,M} \cap \{X_\star > \sqrt{A}\}) \\ &\quad + P(E_{K,M} \cap \{\mathcal{B}_\star + 1 > \sqrt{A}\}). \end{aligned}$$

The limit superior as $K \rightarrow \infty$ of each term tends to zero as $A \rightarrow \infty$. This proves tightness of the products using their marginal tails only; neither independence nor an integrable product has been used.

Density mismatch. On $E_{K,M}$,

$$(\lambda - 1)^2 L = \frac{(\ell - L)^2}{L} \leq Cr_K^2.$$

Consequently (5.15) gives the bound

$$\mathbb{E}_P[\mathbf{1}_{E_{K,M}} (\lambda - 1)^2 L (X_L + X_R)] \leq C_M r_K^2 \rightarrow 0.$$

This particular term vanishes in mean. Combining its vanishing tail with the two tight product terms in (5.13) proves (5.7). $\square$

6. PROOF OF THE DISCREPANCY COMPARISON THEOREM

Let P, Q satisfy Theorem 1.1. Fix r, G and first fix $M < \infty$. Choose r_K in Lemma 4.3 jointly for P and Q , and use this same sequence in both good events $E_{K,M}$.

6.1. Cancellation in the logarithmic derivative. For almost every retained exterior, differentiating the two Gibbs densities gives, for $1 \leq i \leq K$,

$$\partial_i \log \frac{d\mu_\zeta}{d\omega_{K,T}}(u) = \beta B_\zeta(u_i) + \frac{u_i - i}{T}. \tag{6.1}$$

Indeed, the interior pair Hamiltonian is identical in the two laws, and the endpoint charges at 0, L also agree. Both cancel in the density ratio. The remaining exterior-force difference is exactly B_ζ , with the convention of Section 5. The inequality $(a + b)^2 \leq 2a^2 + 2b^2$ therefore gives

$$I(\mu_\zeta | \omega_{K,T}) \leq 2\beta^2 \mathcal{D}_K(\zeta) + \frac{2J_K(\zeta)}{T^2}. \tag{6.2}$$

For each fixed exterior the residual force is bounded on $[0, L]$: the adjacent exterior particles have positive distance from the endpoints, and the discrepancy envelope controls the tails. Thus the logarithm of the density ratio extends to a bounded Lipschitz function on the simplex closure,

and its entropy and Fisher information are finite. This cancellation avoids inverse-gap moment assumptions, including when $0 < \beta < 1$.

6.2. A common confinement scale. Define

$$q_{K,M} = \frac{1}{K^2} \left\{ \mathbb{E}_P[\mathbf{1}_{E_{K,M}} J_K] + \mathbb{E}_Q[\mathbf{1}_{E_{K,M}} J_K] \right\} \longrightarrow 0,$$

where the convergence is Lemma 4.4 for the fixed M . Set

$$\varepsilon_{K,M} = \max\{\sqrt{q_{K,M}}, K^{-1/2}\}, \quad T_{K,M} = \varepsilon_{K,M} K. \quad (6.3)$$

These deterministic sequences may depend on P, Q, M . The same reference $\omega_{K,T_{K,M}}$ is used in both comparisons. In particular,

$$\varepsilon_{K,M} \rightarrow 0, \quad T_{K,M} \geq \sqrt{K} \rightarrow \infty, \quad \frac{T_{K,M}}{K} \rightarrow 0, \quad \frac{q_{K,M}}{\varepsilon_{K,M}} \leq \sqrt{q_{K,M}} \rightarrow 0. \quad (6.4)$$

Thus the confinement becomes weak, while its scale is still small compared with the block length.

Proposition 2.1 and (6.2) imply

$$|\mu_\zeta(F_K) - \omega_{K,T_{K,M}}(F_K)|^2 \leq C_G \left\{ \varepsilon_{K,M} \mathcal{D}_K + \frac{J_K}{\varepsilon_{K,M} K^2} \right\}. \quad (6.5)$$

The two terms are controlled separately. For either candidate law $R \in \{P, Q\}$, Proposition 5.2 and $\varepsilon_{K,M} \rightarrow 0$ imply

$$\forall \delta > 0, \quad R(E_{K,M} \cap \{\varepsilon_{K,M} \mathcal{D}_K > \delta\}) \longrightarrow 0. \quad (6.6)$$

To see the use of tightness explicitly, for every fixed $A > 0$ and all large K with $\varepsilon_{K,M} A \leq \delta$, this probability is at most $R(E_{K,M} \cap \{\mathcal{D}_K > A\})$. Take first the limit superior in K and then let $A \rightarrow \infty$. For the quadratic term, under P ,

$$\mathbb{E}_P \left[\frac{\mathbf{1}_{E_{K,M}} J_K}{\varepsilon_{K,M} K^2} \right] \leq \frac{q_{K,M}}{\varepsilon_{K,M}} \longrightarrow 0,$$

and the same bound holds under Q . Consequently, for $R \in \{P, Q\}$,

$$\mathbb{E}_R \left[\mathbf{1}_{E_{K,M}} \left| \mu_\zeta(F_K) - \omega_{K,T_{K,M}}(F_K) \right| \right] \longrightarrow 0. \quad (6.7)$$

Indeed, the integrand converges to zero in R -probability by (6.5) and is bounded by $2 \|G\|_\infty$. This deduction does not use a first-moment bound on $\mathcal{D}_K$.

6.3. Order of limits and identification. Let $\mathfrak{F}_K(\gamma)$ be the rescaled block statistic $F_K(u(\gamma))$. Conditional DLR and exterior measurability of $E_{K,M}$ imply

$$\limsup_{K \rightarrow \infty} \left| \mathbb{E}_P \mathfrak{F}_K - \omega_{K,T_{K,M}}(F_K) \right| \leq 2 \|G\|_\infty P(\mathfrak{g}_* > M).$$

The analogous estimate holds for Q . For this fixed M , cancelling the common reference gives

$$\limsup_{K \rightarrow \infty} |\mathbb{E}_P \mathfrak{F}_K - \mathbb{E}_Q \mathfrak{F}_K| \leq 2 \|G\|_\infty \{P(\mathfrak{g}_* > M) + Q(\mathfrak{g}_* > M)\}.$$

Only now let $M \rightarrow \infty$. The gap $\mathfrak{g}_*$ is finite almost surely, so

$$\lim_{K \rightarrow \infty} |\mathbb{E}_P \mathfrak{F}_K - \mathbb{E}_Q \mathfrak{F}_K| = 0. \quad (6.8)$$

No estimate uniform in M is needed: the common reference expectation has been eliminated for each fixed M before taking $M \rightarrow \infty$.

The affine rescaling does not alter the limiting statistic. Extending G by zero gives a bounded globally Lipschitz function, and each gap occurs in at most r windows. The unscaled statistic is $\mathfrak{F}_K^{\text{un}} = A_{K,r}$ from (3.10), because $x_i = a_i$. Thus

$$|\mathfrak{F}_K - \mathfrak{F}_K^{\text{un}}| \leq \frac{C_G}{K-r} |c-1| \ell = \frac{C_G}{K-r} |\ell - L| \rightarrow 0$$

almost surely by density one, and also in mean by boundedness. Lemma 3.4 and (6.8) yield

$$\mathbb{E}_{P^0} G(g_0, \dots, g_{r-1}) = \mathbb{E}_{Q^0} G(g_0, \dots, g_{r-1})$$

for every r and $G \in C_c^2((0, \infty)^r)$. These test functions determine probability measures on $(0, \infty)^r$, so every finite consecutive-gap distribution agrees. Point-shift invariance moves any finite integer-indexed gap window into such a consecutive window. It follows that

$$\text{Law}_{P^0}((g_j)_{j \in \mathbb{Z}}) = \text{Law}_{Q^0}((g_j)_{j \in \mathbb{Z}}). \quad (6.9)$$

The Palm configuration is reconstructed from its gaps and the point at zero by

$$p_n = \sum_{j=0}^{n-1} g_j, \quad p_{-n} = - \sum_{j=-n}^{-1} g_j, \quad n \geq 1.$$

Thus $P^0 = Q^0$. Finally Palm inversion at intensity one gives, for bounded measurable f ,

$$\mathbb{E}_P f(\gamma) = \mathbb{E}_{P^0} \left[\int_0^{g_0} f(\theta_t \gamma) dt \right],$$

and the same formula for Q proves $P = Q$. This completes Theorem 1.1, without an ergodicity assumption.

7. FROM LOGARITHMIC ELECTRIC ENERGY TO DISCREPANCY CONTROL

Theorem 1.1 has been proved without an electric-energy assumption. This section verifies its three variance hypotheses for the ergodic finite-energy components used in Theorem 1.2. We separate the imported linear discrepancy estimate from the integrated-variance conclusion proved below.

7.1. Energy convention and results used from the literature. For a configuration $\gamma \in \text{Conf}(\mathbb{R})$, distinguish its embedding in the plane from the one-dimensional counting measure:

$$\bar{\gamma} = \sum_{q \in \gamma} \delta_{(q,0)}, \quad dx \delta_0 = \text{Leb} \otimes \delta_0.$$

The second measure is the unit-density background on the horizontal axis. Integrals over subsets of $\mathbb{R}^2$ use $dx dy$ unless a different measure is displayed. The notation $\langle T, \phi \rangle$ denotes the action of a distribution T on a test function ϕ . Fix $p \in (1, 2)$ and define

$$\text{Comp}(\gamma) = \left\{ E \in L_{\text{loc}}^p(\mathbb{R}^2; \mathbb{R}^2) : -\text{div } E = 2\pi(\bar{\gamma} - dx \delta_0) \text{ in distributions} \right\}. \quad (7.1)$$

Thus the divergence constraint means

$$\int_{\mathbb{R}^2} E \cdot \nabla \phi = 2\pi \left\{ \sum_{q \in \gamma} \phi(q, 0) - \int_{\mathbb{R}} \phi(x, 0) dx \right\} \quad (\phi \in C_c^\infty(\mathbb{R}^2)).$$

No curl-free condition is imposed in this definition.

For $0 < \eta < 1$, let ρ_η be the uniform probability measure on the circle of radius η , characterized by

$$\int \phi(z) \rho_\eta(dz) = \frac{1}{2\pi} \int_0^{2\pi} \phi(\eta \cos \theta, \eta \sin \theta) d\theta.$$

Define, almost everywhere in $z \in \mathbb{R}^2$,

$$f_\eta(z) = \mathbf{1}_{\{|z| \leq \eta\}} \log(|z|/\eta), \quad E_\eta = E + \sum_{q \in \gamma} \nabla f_\eta(\cdot - (q, 0)). \quad (7.2)$$

The value of f_η at zero is irrelevant to this locally integrable function. Since $\Delta f_\eta = 2\pi(\delta_{(0,0)} - \rho_\eta)$, the truncated field $F = E_\eta$ satisfies

$$-\operatorname{div} F = 2\pi(\bar{\gamma} * \rho_\eta - dx \delta_0). \quad (7.3)$$

Define

$$\begin{aligned} \mathcal{W}_\eta(E) &= \limsup_{R \rightarrow \infty} \frac{1}{2R} \int_{[-R, R] \times \mathbb{R}} |E_\eta(x, y)|^2 dx dy + 2\pi \log \eta, \\ \mathcal{W}(E) &= \lim_{\eta \downarrow 0} \mathcal{W}_\eta(E), \\ W(\gamma) &= \inf_{E \in \operatorname{Comp}(\gamma)} \mathcal{W}(E). \end{aligned} \quad (7.4)$$

The spatial average is per unit horizontal length. The order of operations is $R \rightarrow \infty$, then $\eta \downarrow 0$, and finally the infimum over compatible fields. The limits have the extended-real interpretation of the regularized energy, and $\inf \emptyset = +\infty$. The truncation-limit existence is the one recorded in [LS17, Section 2.7.1].

The choice of $p \in (1, 2)$ does not change the finite-energy class. Indeed, if a locally integrable compatible field has finite $\mathcal{W}(E)$, some fixed η has finite spatial mean square energy. Then $E_\eta \in L^2_{\text{loc}}$, and on each compact set $E - E_\eta$ is a finite sum of truncated Coulomb singularities. Each such gradient belongs to L^p_{loc} for every $p < 2$. Hence every field relevant to a finite infimum in (7.4) belongs to the displayed admissible class.

Lemma 7.1 (Energy normalization and the linear variance bound). *Write $W_{\text{LS}}(\gamma, 1)$ and $\widetilde{W}_{\text{LS}}(P, 1)$ for the configuration and process energies of [LS17, Sections 2.7.3–2.7.4] in the Log1 case. Then*

$$W(\gamma) = 2\pi W_{\text{LS}}(\gamma, 1), \quad \mathbb{E}_P W = 2\pi \widetilde{W}_{\text{LS}}(P, 1). \quad (7.5)$$

Writing $\widetilde{\mathcal{W}}^{\text{div}}$ and $\mathcal{W}^{\text{div}}$ for the configuration and process energies of [Ass26, Definitions 2.5–2.6], one also has

$$W(\gamma) = 2\pi \widetilde{\mathcal{W}}^{\text{div}}(\gamma), \quad \mathbb{E}_P W = 2\pi \mathcal{W}^{\text{div}}(P). \quad (7.6)$$

Thus the finite-energy hypothesis in Theorem 1.2 is exactly the energy hypothesis of [Ass26, Theorem 1.1]. In particular, there is an absolute constant $C_0 < \infty$ such that $W(\gamma) \geq -C_0$. If P is stationary and $\mathbb{E}_P W < \infty$, then

$$\mathbb{E}_P N_A = \operatorname{Leb}(A), \quad v_P(t) \leq C_P t \quad (t > 0), \quad (7.7)$$

for every bounded Borel set A , with $C_P < \infty$ depending on P .

Proof. Specialize [LS17, Sections 2.6.4–2.7.4] to $d = 1$, extension dimension $k = 1$, weight exponent zero, background $m = 1$, and $c_{d,s} = 2\pi$. The admissible field class and divergence equation then agree with ours. Their positive cutoff is

$$f_\eta^{\text{LS}}(z) = (-\log |z| + \log \eta)_+ = -f_\eta(z),$$

and their truncation $E - \sum \nabla f_\eta^{\text{LS}}$ is exactly E_η . Their cube $\square_S$ has side length S ; taking $S = 2R$ identifies their spatial average with ours. Finally their subtraction $-c_{d,s}g(\eta)$ is $+2\pi \log \eta$, and their configuration energy includes a factor $1/c_{d,s}$. This proves (7.5). The lower bound is the standard one recalled in the proof of [LS17, Lemma 3.9].

For (7.6), take the exponent $q_{\text{el}} = p$ in [Ass26, Definition 2.5]. Its field class has exactly the same divergence constraint, with no curl-free requirement. Its positive inner cutoff is f_η^{LS} , so the

regularized fields agree. Definition 2.6 there uses the same strip, order of limits, and infimum, with regularized functional

$$\frac{1}{2R2\pi} \int_{[-R,R] \times \mathbb{R}} |E_\eta|^2 + \log \eta.$$

This is our entire regularized functional divided by 2π . Positive scaling commutes with the limits and infimum, giving the configuration identity. The process energy in that definition is the expectation of this configuration infimum, so the process identity follows directly; no optimization over stationary joint distributions of configurations and fields, and no measurable field selection, is involved.

By [LS17, Lemma 3.2] and (7.5), for $t \geq 1$,

$$\mathbb{E}_P(N_{(0,t]} - t)^2 \leq C(C + \mathbb{E}_P W)t.$$

Stationarity forces intensity one, since any different intensity would give a quadratic lower bound by Jensen's inequality. For completeness, the small-interval bound follows from stationarity alone once $\mathbb{E}_P N_{(0,1]}^2 < \infty$. If $0 < t < 1$ and $m = \lfloor 1/t \rfloor$, then

$$m\mathbb{E}_P N_{(0,t]}^2 = \mathbb{E}_P \sum_{j=0}^{m-1} N_{(jt, (j+1)t]}^2 \leq \mathbb{E}_P N_{(0,1]}^2.$$

Since $m \geq 1/(2t)$, this gives $v_P(t) \leq 2t\mathbb{E}_P N_{(0,1]}^2$ and completes (1.6). $\square$

Remark 7.2 (Truncation versus normalization). The cutoff displayed in [DHL19, Definition 2.20], in the arXiv version specified in the bibliography, has $|z| \geq \eta$. We use the compatible underlying formulation of [LS17], matched explicitly in Lemma 7.1. The singular energy of a unit charge is

$$\int_{\eta < |z| < r_0} |\nabla \log |z||^2 dz = 2\pi \log(r_0/\eta).$$

Thus $1/(2R)$ over $[-R, R] \times \mathbb{R}$ pairs with $+2\pi \log \eta$. Using $1/R$ over that window would require $+4\pi \log \eta$. Multiplying the entire regularized functional, including its subtraction, by a positive constant preserves finiteness. Changing the cutoff or only one of those two coefficients has no such justification. The identity (7.5) documents the applicability of the imported energy results.

7.2. A stationary configuration–field law and the spectral estimate. For vector fields, the horizontal translation is $(\theta_a F)(x, y) = F(x + a, y)$. A joint law of (γ, F) is called stationary when it is invariant under $(\gamma, F) \mapsto (\theta_a \gamma, \theta_a F)$ for every $a \in \mathbb{R}$.

Lemma 7.3 (Stationary joint law of a configuration and a field). *Let P be ergodic and stationary, and suppose $P(W < \infty) = 1$, where W is defined in (7.4). There exist $\eta \in (0, 1)$ and a probability measure $\mathbf{Q}$ on configuration–field pairs such that*

$$\begin{aligned} \text{Law}_{\mathbf{Q}}(\gamma) &= P, \\ (\theta_a, \theta_a)_\# \mathbf{Q} &= \mathbf{Q} \quad (a \in \mathbb{R}), \\ -\text{div } F &= 2\pi(\bar{\gamma} * \rho_\eta - dx \delta_0) \quad \mathbf{Q}\text{-almost surely}, \end{aligned} \tag{7.8}$$

and

$$\mathcal{E} = \mathbb{E}_{\mathbf{Q}} \int_{[0,1] \times \mathbb{R}} |F(x, y)|^2 dx dy < \infty. \tag{7.9}$$

Proof. Choose a configuration γ with $W(\gamma) < \infty$ such that

$$\lim_{R \rightarrow \infty} \frac{1}{2R} \int_{-R}^R f_j(\theta_x \gamma) dx = P(f_j), \quad j \geq 1, \tag{7.10}$$

where $(f_j)_{j \geq 1}$ is a countable convergence-determining family of bounded continuous functions on the space of locally finite measures with the vague topology. Ergodicity ensures that such configurations have full probability. Choose a compatible field whose renormalized functional is finite; attainment

of the infimum defining W is unnecessary. At some sufficiently small fixed η , the truncated field $F = E_\eta$ consequently satisfies

$$A = \limsup_{R \rightarrow \infty} \frac{1}{2R} \int_{[-R, R] \times \mathbb{R}} |F|^2 < \infty.$$

Average translates of this single pair:

$$\mathbf{Q}_R = \frac{1}{2R} \int_{-R}^R \delta_{(\theta_x \gamma, \theta_x F)} dx.$$

No field is selected as a measurable function of the configuration.

For the field coordinate use $H_{\text{loc}}^{-1}(\mathbb{R}^2; \mathbb{R}^2)$: a distribution belongs to this space if its product with every smooth compactly supported cutoff belongs to $H^{-1}(\mathbb{R}^2; \mathbb{R}^2)$, the dual of $H^1(\mathbb{R}^2; \mathbb{R}^2)$. The local topology is generated by these cutoff norms on an exhausting sequence of bounded domains. For every $a, b > 0$, Fubini gives

$$\mathbb{E}_{\mathbf{Q}_R} \int_{[-a, a] \times [-b, b]} |F|^2 \leq \frac{2a}{2R} \int_{[-R-a, R+a] \times \mathbb{R}} |F|^2, \quad \limsup_{R \rightarrow \infty} \mathbb{E}_{\mathbf{Q}_R} \int_{[-a, a] \times [-b, b]} |F|^2 \leq 2aA.$$

Markov's inequality on an exhausting sequence of boxes and the compact embedding of bounded local L^2 sets into H_{loc}^{-1} give tightness of the field laws. The configuration marginals converge to P , hence are tight as well. A subsequence therefore has a joint weak limit $\mathbf{Q}$, whose configuration marginal is exactly P . The local L^2 norm, extended to $+\infty$ outside L^2 , is lower semicontinuous for distributional convergence. The same bounds show that $\mathbf{Q}$ is supported on locally square-integrable fields.

For each fixed horizontal shift h , translating the averaging interval changes $\mathbf{Q}_R$ in total variation by at most $|h|/R$. Translations are continuous in both coordinate topologies, so $\mathbf{Q}$ is stationary. For any $u \in C_c^\infty(\mathbb{R}^2)$, the maps

$$F \longmapsto \langle F, \nabla u \rangle, \quad \gamma \longmapsto \int \left(\int u(p + z_1, z_2) \rho_\eta(dz) \right) \gamma(dp)$$

are continuous in these topologies. The inner integral in the second map is a continuous compactly supported function of p . Thus (7.3) passes to the joint limit, first on a countable dense family of tests and then on all tests. Finally the analogous averaging bound for $(0, 1) \times (-b, b)$ has limsup at most A . Lower semicontinuity followed by monotone convergence as $b \rightarrow \infty$ proves $\mathcal{E} \leq A$ in (7.9). $\square$

In the rest of this subsection, an expectation involving F is under the fixed joint law $\mathbf{Q}$; for configuration observables it is under its marginal P . We isolate integration by parts with a configuration-dependent test. For a stationary process of intensity one with finite local second moments, let $\sigma = \gamma - \text{Leb}$ be the centered random measure on $\mathbb{R}$. Its covariance spectral measure, denoted by S_P (not the relative entropy $S(\mu | \omega)$), is normalized by

$$\mathbb{E}_P |\langle \sigma, f \rangle|^2 = \int_{\mathbb{R}} |\hat{f}(k)|^2 S(dk), \quad \hat{f}(k) = \int_{\mathbb{R}} e^{-ikx} f(x) dx. \quad (7.11)$$

The identity is first understood for smooth compactly supported test functions f . Here S_P is a symmetric positive Radon measure and need not have a density. Finite local second moments make σ a stationary random tempered distribution, so this definition also applies to Schwartz tests by approximation.

Lemma 7.4 (Stationary random-test integration by parts). *Suppose P has intensity one and finite local second moments, and (γ, F) has a stationary joint law $\mathbf{Q}$ satisfying (7.3) and (7.9) at fixed η . For real even $\chi \in C_c^\infty(\mathbb{R} \setminus \{0\})$, define*

$$h_y(x) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{ikx} \chi(k) e^{-|k||y|} dk, \quad \varphi(x, y) = \langle \sigma, h_y(x - \cdot) \rangle.$$

Then

$$\mathbb{E} \int_{[0,1] \times \mathbb{R}} |\nabla \varphi|^2 = 2 \int_{\mathbb{R}} |k| \chi(k)^2 S(dk), \quad (7.12)$$

and

$$\mathbb{E} \int_{[0,1] \times \mathbb{R}} F \cdot \nabla \varphi = 2\pi \int_{\mathbb{R}} \chi(k) a_\eta(k) S_P(dk), \quad (7.13)$$

where

$$a_\eta(k) = \frac{1}{2\pi} \int_0^{2\pi} e^{ik\eta \cos \theta - |k|\eta |\sin \theta|} d\theta.$$

Proof. Fix $0 < c < C < \infty$ such that $\text{supp } \chi \subset \{c \leq |k| \leq C\}$. The kernels h_y and their horizontal derivatives are Schwartz functions, with each Schwartz seminorm bounded by a constant times $e^{-c|y|/2}$; polynomial factors arising from spectral differentiation are absorbed in this exponential bound. The same holds for their first vertical derivatives on each side of $y = 0$. Local point counts grow at most polynomially almost surely, by stationarity and the first moment bound. Thus the convolutions define pathwise smooth functions away from $y = 0$, continuous across $y = 0$, with a weak first vertical derivative. Their mean is zero. Applying (7.11) at each y and integrating $2k^2 e^{-2|k||y|}$ over $\mathbb{R}$ gives (7.12).

Here are explicit cutoffs for (7.13). Let q_ϵ be a nonnegative even smooth mollifier in the vertical variable, supported in $[-\epsilon, \epsilon]$, with mass one, and set $\varphi_\epsilon = q_\epsilon *_y \varphi$, $0 < \epsilon < 1$. It is smooth in both variables. The compact spectral support gives, with constants depending on χ and S_P on its support,

$$\mathbb{E}(|\varphi_\epsilon(0, y)|^2 + |\nabla \varphi_\epsilon(0, y)|^2) \leq C_\chi e^{-c|y|}, \quad (7.14)$$

$$\mathbb{E} \int_{\mathbb{R}} |\nabla \varphi_\epsilon(0, y) - \nabla \varphi(0, y)|^2 dy \leq C_\chi \epsilon. \quad (7.15)$$

For the latter estimate, outside $|y| \leq \epsilon$ the multipliers and their first derivatives change by $O(\epsilon) e^{-c|y|}$; inside that strip the first derivatives are uniformly bounded and the strip has length 2ϵ . This accounts for the cusp without assuming pointwise values of F on the line.

Take $w \in C_c^\infty(\mathbb{R})$ nonnegative with $\int w = 1$, and put $w_L(x) = L^{-1} w(x/L)$. Take $\vartheta \in C_c^\infty(\mathbb{R})$ equal to one on $[-1, 1]$ and supported in $[-2, 2]$, and put $\vartheta_H(y) = \vartheta(y/H)$, where $H > \max(1, \eta)$. The distribution identity can be tested pathwise against $w_L \vartheta_H \varphi_\epsilon$: it holds simultaneously for all smooth compactly supported tests on a single event of probability one. Consequently the identity also holds for this test after the configuration has been fixed. Every expected pairing is integrable. On the field side this follows from (7.9) and (7.14); on the charge side it follows from local second moments and the Schwartz bounds. In particular no third moment of the counts is used.

Expand the gradient of this product and take expectation. Stationarity makes the main term independent of L , because $\int w_L = 1$. Cauchy–Schwarz and $\int |w'_L| = L^{-1} \int |w'|$ bound the horizontal cutoff error by

$$\left| \mathbb{E} \int_{\mathbb{R}^2} w'_L \vartheta_H F_x \varphi_\epsilon \right| \leq \frac{C_\chi \sqrt{\mathcal{E}}}{L}.$$

The vertical cutoff error and the omitted tail of the main term are bounded, respectively, by

$$\left| \mathbb{E} \int_{\mathbb{R}^2} w_L \vartheta'_H F_y \varphi_\epsilon \right| \leq \frac{C_\chi \sqrt{\mathcal{E}}}{H} e^{-cH/2}, \quad C_\chi \sqrt{\mathcal{E}} e^{-cH/2}.$$

These constants are uniform in L, H and $0 < \epsilon < 1$.

For clarity, the expected charge pairing can be computed before removing either spatial cutoff. Put

$$b_\epsilon(k, y) = \int q_\epsilon(s) e^{-|k||y-s|} ds.$$

The covariance identity and stationarity give, as an identity of measures in p ,

$$\mathbb{E}[\gamma(dp) \varphi_\epsilon(p+a, b)] = \left(\int e^{ika} \chi(k) b_\epsilon(k, b) S_P(dk) \right) dp.$$

It follows first by testing against a smooth compactly supported function of p and then by the Schwartz approximation above. Since $\vartheta_H = 1$ on the support of ρ_η and $\int w_L(p+a) dp = 1$, the right side of the tested divergence identity is exactly

$$2\pi \int \chi(k) a_{\eta, \epsilon}(k) S_P(dk), \quad a_{\eta, \epsilon}(k) = \int e^{ikz_1} b_\epsilon(k, z_2) \rho_\eta(dz).$$

The original line background has zero expected pairing since $\mathbb{E}\varphi_\epsilon = 0$. Thus there is no extra contribution from the difference between the original and smeared backgrounds.

Let $L \rightarrow \infty$, then $H \rightarrow \infty$, and finally $\epsilon \downarrow 0$, in that order. The first two limits follow from the displayed cutoff bounds. By (7.15) the field pairing changes in the last limit by at most $C_\chi \sqrt{\mathcal{E}\epsilon}$. On $\text{supp } \chi$, $|a_{\eta, \epsilon} - a_\eta| \leq C\epsilon$; the charge pairing therefore converges as well, since S_P is finite on that compact set. This proves (7.13). $\square$

Proposition 7.5 (Spectral energy and hyperuniformity). *Suppose P is stationary of intensity one, satisfies (1.6), and admits a stationary joint law with an electric field as in Lemma 7.3. Then*

$$\lim_{t \rightarrow \infty} \frac{v_P(t)}{t} = 0, \quad \int_1^\infty \frac{v_P(t)}{t^2} dt < \infty. \quad (7.16)$$

Thus both (1.7) and (1.8) hold.

Proof. The linear variance bound gives finite local second moments, so define S_P by (7.11). Approximation of interval indicators by smooth tests gives

$$v_P(t) = \int_{\mathbb{R}} \frac{4 \sin^2(kt/2)}{k^2} S_P(dk), \quad (7.17)$$

with the continuous value t^2 at zero. In particular, $t^2 S(\{0\}) \leq v_P(t) \leq C_P t$ for $t \geq 1$, hence $S_P(\{0\}) = 0$.

Apply Lemma 7.4 and Cauchy–Schwarz to obtain

$$\left| \int_{\mathbb{R}} \chi(k) a_\eta(k) S_P(dk) \right|^2 \leq \frac{\mathcal{E}}{2\pi^2} \int_{\mathbb{R}} |k| \chi(k)^2 S_P(dk). \quad (7.18)$$

The coefficient is obtained by dividing the product of (7.9) and (7.12) by $(2\pi)^2$. The function a_η is real and even, is smooth away from zero, and is continuous with $a_\eta(0) = 1$. Choose $\delta \in (0, 1)$ so that $a_\eta(k) \geq 1/2$ whenever $|k| < \delta$. For a smooth even cutoff $0 \leq \psi \leq 1$ supported in $\epsilon < |k| < \delta$, the test $\chi(k) = a_\eta(k) \psi(k)/|k|$ belongs to $C_c^\infty(\mathbb{R} \setminus \{0\})$: every integral is finite on this compact annulus before any limit is taken. Write

$$A_\psi = \int \frac{a_\eta(k)^2}{|k|} \psi(k) S_P(dk).$$

The rightmost integral in (7.18) is at most A_ψ , because $\psi^2 \leq \psi$. Therefore

$$A_\psi^2 \leq \frac{\mathcal{E}}{2\pi^2} A_\psi, \quad A_\psi \leq \frac{\mathcal{E}}{2\pi^2}.$$

Exhaust the punctured interval by such cutoffs and use monotone convergence and $a_\eta^2 \geq 1/4$ to deduce

$$\int_{0 < |k| < \delta} \frac{S_P(dk)}{|k|} \leq \frac{2\mathcal{E}}{\pi^2} < \infty. \quad (7.19)$$

The annular tests establish this integrability; it was not assumed in constructing the potential or applying integration by parts.

For high frequencies, integrate (7.17) over $1 \leq t \leq 2$. The integral of $v_P(t)$ is finite. The continuous function $\int_1^2 \sin^2(kt/2) dt$ is strictly positive on every compact annulus and tends to $1/2$ as $|k| \rightarrow \infty$; it therefore has a positive lower bound on $|k| \geq \delta$. Tonelli yields

$$\int_{|k| \geq \delta} k^{-2} \mathbf{S}_P(dk) < \infty. \quad (7.20)$$

Another application of Tonelli, now using

$$\frac{4}{k^2} \int_1^\infty \frac{\sin^2(kt/2)}{t^2} dt \leq \begin{cases} C/|k|, & 0 < |k| < \delta, \\ 4/k^2, & |k| \geq \delta, \end{cases}$$

proves (1.8). Finally, for $t \geq 1$ the integrand $4\sin^2(kt/2)/(tk^2)$ of $v_P(t)/t$ tends to zero at every $k \neq 0$. On $0 < |k| < \delta$ it is bounded by $C/|k|$, using $\sin^2 u \leq \min(u^2, 1)$; on $|k| \geq \delta$ it is bounded by $4/k^2$. Equations (7.19) and (7.20), and the absence of an atom at zero, permit dominated convergence. This proves (1.7). $\square$

7.3. The finite-energy implication.

Proposition 7.6 (Variance hypotheses for finite-energy components). *Let P be an ergodic stationary canonical logarithmic DLR solution with $\mathbb{E}_P W < \infty$, where W is given by (7.4). Then P has intensity one, and there is $C_P < \infty$ such that*

$$v_P(t) \leq C_P t \quad (t > 0), \quad \lim_{t \rightarrow \infty} \frac{v_P(t)}{t} = 0, \quad \int_1^\infty \frac{v_P(t)}{t^2} dt < \infty. \quad (7.21)$$

These are exactly (1.6)–(1.8); C_P may depend on P .

Proof. Lemma 7.1 gives intensity one and the linear variance bound. Because W is bounded below, finite expectation implies $W < \infty$ almost surely. Lemma 7.3 constructs the stationary joint distribution of the configuration and field; Proposition 7.5 then proves the sublinear and integrated variance bounds. In particular, the integrated bound is a conclusion of the spectral argument, not an imported consequence of the linear discrepancy estimate alone. $\square$

Proof of Theorem 1.2. By the canonical DLR ergodic decomposition in [DHL19, Proposition 3.13], there are a probability space $(\mathcal{A}, \pi)$ and ergodic stationary canonical DLR solutions P_α such that

$$P(f) = \int_{\mathcal{A}} P_\alpha(f) \pi(d\alpha) \quad \text{for every bounded measurable } f. \quad (7.22)$$

Here the DLR specification is the conditional-density specification described in Section 1; its preservation under this decomposition uses the probabilistic argument of that proposition, independently of any numerical convention for the energy. With C_0 as in Lemma 7.1, Tonelli's theorem gives

$$\int_{\mathcal{A}} \mathbb{E}_{P_\alpha}(W + C_0) \pi(d\alpha) = \mathbb{E}_P(W + C_0) < \infty.$$

Hence $\mathbb{E}_{P_\alpha} W < \infty$ for π -almost every α . Proposition 7.6 puts each such component in the class of Theorem 1.1. Hence all these components coincide with a single law.

The two existence inputs are used separately. The canonical DLR property of Sine_β is [DHL19, Theorem 2.1]. Its finite expected energy follows from [LS17, Corollary 1.2]: Sine_β minimizes a free energy with finite minimum, and the specific relative entropy in that functional is nonnegative. Equation (7.5) converts this statement to $\mathbb{E}_{\text{Sine}_\beta} W < \infty$ in our convention. Apply the same ergodic decomposition to Sine_β . All its components belong to the singleton class of Theorem 1.1, so their common law is Sine_β . Every candidate P , being a mixture of this law, equals Sine_β . $\square$

8. RELATION TO OTHER UNIQUENESS ARGUMENTS

The finite-energy scope of Theorem 1.2 agrees with [Ass26, Theorem 1.1], by (7.6) and the DLR comparison below. Both results allow nonergodic stationary laws. The two proofs share a passage from entropy to averaged gap statistics and then to Palm identification. More specifically, [Ass26, Proposition 7.3] obtains an exterior-averaged relative entropy of order $o(L)$ against the matched circular β ensemble. Cyclic-gap transport then distributes that cost among the gaps [Ass26, Lemma 8.3], and the circular ensemble's local and Palm limits identify the process [Ass26, Sections 8.3–8.6]. The present proof uses the static Brascamp–Lieb and logarithmic Sobolev estimate of Proposition 2.1, together with force tightness and weak quadratic confinement. It compares both candidates with one lattice-boundary law and cancels the common reference. The linear and sublinear variance bounds are the discrepancy inputs recorded in [Ass26, Proposition 2.7]; the integrated variance condition (1.8) is the additional explicit sufficient input used by our force-envelope argument and proved here from energy.

Remark 8.1 (Agreement of the DLR conventions). On the almost-sure regularity event of Lemma 3.1, the interval formulation used here extends to the bounded Borel resampling sets in [Ass26, Definitions 2.1–2.2]. To check its uniformity requirement, fix a bounded Borel set $\Lambda \subset [-B, B]$ with $B > 0$ and $|\Lambda| > 0$, put $n = N_\Lambda(\gamma)$, and define

$$\text{Conf}_n(\Lambda) = \{\eta \in \text{Conf}(\mathbb{R}) : \eta(\Lambda) = n, \eta(\Lambda^c) = 0\}.$$

For $\eta \in \text{Conf}_n(\Lambda)$, write

$$M_{\Lambda,R}(\eta, \gamma) = - \sum_{\substack{p \in \gamma \setminus \Lambda \\ |p| \leq R}} \int_{\Lambda} \log |x - p| (\eta - \gamma_\Lambda)(dx)$$

for the relative exterior interaction difference at radius R . For $S > R > 2B$, Taylor expansion and cancellation of the total mass give, uniformly over all n -point configurations η in Λ ,

$$|M_{\Lambda,S}(\eta, \gamma) - M_{\Lambda,R}(\eta, \gamma)| \leq 2nB \left| \sum_{\substack{p \in \gamma \\ R < |p| \leq S}} \frac{1}{p} \right| + CnB^2 \sum_{\substack{p \in \gamma \\ |p| > R}} \frac{1}{p^2}.$$

Density one makes the second series summable, and the reciprocal principal value makes the first term Cauchy. Thus the relative interaction differences converge uniformly on the set of all n -point configurations in Λ , including along the radii $m/2$, $m \in \mathbb{N}$, used in [Ass26].

Choose a bounded interval containing Λ and condition its canonical Gibbs law further on the points outside Λ and on N_Λ . Disintegration gives precisely the kernel of [Ass26, Definition 2.1]. Its normalizer is positive and finite: the distant potential is bounded after fixing its additive constant, while the finitely many nearby logarithmic interaction factors are bounded above and positive Lebesgue-almost everywhere on the bounded domain. The binomial reference measure is the law of n independent points uniformly distributed in Λ . Its ordered-coordinate density is $n!/\text{Leb}(\Lambda)^n$, a constant which cancels in normalization. Subtracting the interaction difference to a fixed reference configuration with the same exterior and particle number removes dependence on the original interior configuration and gives exterior-and-count measurability. Limits of the finite-cutoff kernels, or a conditional-law version with a countable determining class, give a measurable version of the conditional kernel, up to null sets, as required by the formulation of [Ass26, Definition 2.2]. For particle number zero, the kernel is the point mass on the empty configuration. Conversely, that definition immediately includes interval resampling. Assiotis initially allows multiplicities, but [Ass26, Lemma 2.8] makes every canonical DLR law simple.

For the finite-energy class, the required regularity holds on almost every ergodic component by Proposition 7.6 and Lemma 3.1, hence also under their mixture. This verifies agreement of the DLR classes used in the two finite-energy statements.

At $\beta = 2$, the deterministic theorem of [KMD19, Theorem 1.3] provides another route. The reciprocal principal value and density-one conditions needed there follow from the discrepancy bounds, so its kernel convergence applies to every admissible candidate boundary. DLR and convergence of compactly supported Laplace functionals then identify the process.

For general β , the main theorem of the arXiv version of [BEY12] specified in the bibliography concerns a fixed real analytic uniformly convex potential on the line, and its local comparison is combined with strong particle-location estimates. A direct application to arbitrary random DLR exterior potentials would require checking assumptions not provided by macroscopic potential convergence alone. The present proof uses the static convexity inequalities from that framework and compares with one common reference law. In particular, it establishes neither a separate universality theorem for all deterministic boundaries nor a limiting theorem for ω_{K,T_K} .

The distinction between the two improved discrepancy inputs is useful: $v_P(t) = o(t)$ makes the average quadratic displacement $o(K)$ per particle, whereas $\int_1^\infty v_P(t)t^{-2}dt < \infty$ supplies the half-line force envelopes. The linear bound alone gives neither of these two improvements. The weak confinement can be chosen between the quadratic displacement scale and the block length precisely because the improvement is little- o ; no fixed power rate is needed.

APPENDIX A. CONVEX INEQUALITIES ON THE SINGULAR SIMPLEX

We record the approximation used in Proposition 2.1. Let ν be a probability measure with a positive continuous Lebesgue density inside Ω_K . Define its weighted weak Sobolev space by

$$H^1(\nu) = \left\{ f \in H_{\text{loc}}^1(\Omega_K) : \int_{\Omega_K} (f^2 + |\nabla f|^2) d\nu < \infty \right\}. \quad (\text{A.1})$$

Here ∇f is the distributional weak gradient. This definition imposes no boundary trace condition. For a nonnegative function h , write

$$\text{Ent}_\nu(h) = \int h \log h d\nu - \left(\int h d\nu \right) \log \left(\int h d\nu \right),$$

with the usual extended-value convention.

Lemma A.1 (Removal of the gap truncation). *Suppose $V \in C^2(\Omega_K)$, $0 < Z = \int_{\Omega_K} e^{-V} < \infty$, and $V'' \geq \rho \text{Id}_K$ for some $\rho > 0$. Put $\nu = Z^{-1}e^{-V} du$. For every $f \in H^1(\nu)$,*

$$\text{Var}_\nu f \leq \int \nabla f^\top (V'')^{-1} \nabla f d\nu, \quad (\text{A.2})$$

$$\text{Ent}_\nu(f^2) \leq \frac{2}{\rho} \int |\nabla f|^2 d\nu \quad (f \in H^1(\nu)). \quad (\text{A.3})$$

In particular, for $\mu = h\nu$ with $\sqrt{h} \in H^1(\nu)$, $S(\mu | \nu) \leq I(\mu | \nu)/(2\rho)$, where $I = 4 \int |\nabla \sqrt{h}|^2 d\nu$. These statements apply to $V = \Phi_{K,T} - sF_K$ throughout the tilt range in Section 2, with the Hessian bounds stated there and constants unaffected by removal of the truncation.

Proof. For $0 < \delta < 1$ define the nonempty convex simplex

$$\Omega_{K,\delta} = \{u : g_i(u) > \delta \text{ for } 0 \leq i \leq K\}, \quad \nu_\delta = \frac{\mathbf{1}_{\Omega_{K,\delta}} \nu}{\nu(\Omega_{K,\delta})}.$$

Its closure lies strictly inside Ω_K , so the density there is bounded above and below by positive constants and V is C^2 on a neighborhood of the closure. The bounds may depend on δ ; the convexity constant ρ does not. When applying a smooth-domain argument below, one may first mollify V on

this neighborhood. Mollification preserves $V'' \geq \rho \text{Id}_K$, and convergence in C^2 on the closure allows passage back to the stated C^2 potential.

We specify the convex-domain inequalities being used. For a smooth bounded convex domain D and $\nu_D \propto e^{-V} \mathbf{1}_D$, the Brascamp–Lieb and Bakry–Émery inequalities have precisely the forms (A.2)–(A.3), with D as the integration domain; see [BL76, BE85] for the underlying inequalities. Their boundary extension can be seen directly with the Neumann generator $\mathcal{L} = \Delta - \nabla V \cdot \nabla$. Let n denote the outward unit normal on ∂D . If $-\mathcal{L}w = f - \nu_D(f)$ and $\partial_n w = 0$, the integrated Bochner identity and convexity of ∂D give

$$\int (\mathcal{L}w)^2 d\nu_D \geq \int \nabla w^\top V'' \nabla w d\nu_D.$$

The omitted nonnegative terms are $\int \|D^2 w\|_{\text{HS}}^2 d\nu_D$ and the boundary integral of the second fundamental form evaluated on the tangential gradient. Here $\|D^2 w\|_{\text{HS}}^2 = \sum_{i,j} |\partial_{ij} w|^2$ is the squared Hilbert–Schmidt norm of the Hessian. Integration by parts and weighted Cauchy–Schwarz then give

$$\text{Var}_{\nu_D} f = \int \nabla f \cdot \nabla w d\nu_D \leq \left(\int \nabla f^\top (V'')^{-1} \nabla f d\nu_D \right)^{1/2} (\text{Var}_{\nu_D} f)^{1/2},$$

which proves the variance statement. For the Neumann semigroup P_t , entropy dissipation for positive h and the same Bochner identity yield

$$-\frac{d}{dt} \text{Ent}_{\nu_D}(P_t h) = \mathcal{J}_D(P_t h), \quad \frac{d}{dt} \mathcal{J}_D(P_t h) \leq -2\rho \mathcal{J}_D(P_t h), \quad \mathcal{J}_D(h) = \int \frac{|\nabla h|^2}{h} d\nu_D.$$

Integrating to equilibrium gives $\text{Ent}_{\nu_D}(h) \leq \mathcal{J}_D(h)/(2\rho)$. Apply this to $h = f^2 + \varepsilon$ and let $\varepsilon \downarrow 0$ to obtain the stated factor $2/\rho$. Smooth approximation on a smooth bounded domain extends both inequalities to its ordinary H^1 space. This step uses only weights bounded above and below, and makes no assertion about density at collision faces.

Corners of $\Omega_{K,\delta}$ can be removed without changing the potential or its Hessian: the sublevel domains

$$D_m = \left\{ u \in \Omega_{K,\delta} : \sum_{i=0}^K \frac{1}{g_i(u) - \delta} < m \right\}$$

are smooth bounded convex domains for m above the minimum of the displayed strictly convex function, and exhaust $\Omega_{K,\delta}$. Applying the preceding inequalities on D_m and passing to the limit gives them for ν_δ . Every fixed $f \in H^1(\nu)$ restricts to ordinary H^1 on these domains and on $\Omega_{K,\delta}$, by local comparability of the weights. Variance passes by convergence of its two moments, and the right sides pass by monotone convergence of the unnormalized integrals. The entropy passage is justified by the exact formula below, used first with D_m and then with $\Omega_{K,\delta}$.

Indeed, set $z_\delta = \nu(\Omega_{K,\delta})$ and $a_\delta = \int_{\Omega_{K,\delta}} f^2 d\nu$. Then

$$z_\delta \text{Ent}_{\nu_\delta}(f^2) = \int_{\Omega_{K,\delta}} f^2 \log f^2 d\nu - a_\delta \log(a_\delta/z_\delta).$$

As $\delta \downarrow 0$, $z_\delta \rightarrow 1$ and $a_\delta \rightarrow \nu(f^2)$. The negative part of $f^2 \log f^2$ is bounded by $1/e$, while its positive part passes by monotone convergence. Thus the left side converges to $\text{Ent}_\nu(f^2)$, even if initially that quantity is infinite. Multiplying the truncated logarithmic Sobolev inequality by z_δ and taking the limit proves (A.3) for the full weak Sobolev domain directly. The same restriction argument proves (A.2).

For $V = \Phi_{K,T} - sF_K$, the normalizing integral is finite because F_K is bounded. Throughout $|s| \leq c_G K$ its Hessian satisfies both $V'' \geq \frac{1}{2} \Phi''_{K,T}$ and $V'' \geq T^{-1} \text{Id}_K$ on every truncated domain. In particular, the pointwise bound $\nabla F_K^\top (V'')^{-1} \nabla F_K \leq 2A_1/K$ survives the limit unchanged. Taking $s = 0$, $\rho = T^{-1}$ and $f = \sqrt{h}$ proves (2.13) in exactly the Fisher-information convention used in

Section 2. All integrations by parts took place at strictly positive gaps. No inverse-gap integrability or boundary trace at a collision is required, for any $\beta > 0$. $\square$

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